



**The 876 Study:
Integrated Biological and Behavioral Surveillance Survey with Population Size Estimation
Among Men who have Sex with Men and Transgender Persons in Jamaica**



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Executive Summary

Jamaica's National Integrated Strategic Plan (NISP) 2014-2019 (Ministry of Health [MOH] 2015) highlights the need for programs directly targeted to key populations including men who have sex with men (MSM) and transgender persons (TGP). Concurrently, the plan recognizes the need for more data on MSM and TGP to accurately capture the current HIV burden among these populations. The 876 Study was the third cross-sectional study among MSM and TGP in relation to sexual behavior and HIV prevalence. The specific objectives were: a) to determine the prevalence of HIV and its associated risk factors; b) assess engagement along the continuum of HIV prevention and care among MSM and TGP; c) estimate the population size of MSM and TGP; d) gauge the psychosocial needs of MSM and TGP; e) recruit a cohort for future HIV incidence surveillance and engagement in care; f) build local capacity to conduct BSS and population size estimation among key populations in Jamaica; and g) strengthen syphilis reporting, referral, and linkage to treatment services.

The 876 Study used respondent driven sampling (RDS) to recruit MSM and TGP across Jamaica for a bio-behavioral surveillance survey (BSS). RDS leverages social networks by having participants recruit peers known to be part of the population. The study comprised every parish of Jamaica, with study offices located in Kingston, Montego Bay, and Ocho Rios.

MSM and TGP were able to participate if they were aged 16 years or older, biologically male at birth or transgender men (TGM) or transgender women (TGW) by self-identity, had anal or oral sex with a man in the year preceding the study, resided in Jamaica at the time of the study, were able to provide informed consent, and had not previously participated. Between December 2016 and September 2017, the 876 Study recruited 537 self-identified MSM, 102 TGW, and 13 TGM. This report presents findings for the sample as a whole (MSM + TGP) and for MSM and TGW separately where indicated. Too few TGM were recruited for separate analysis.

HIV prevalence in the combined population (MSM + TGP) was 32.9% (95% confidence interval [CI]: 28.2%-37.7%). A higher proportion of TGW were HIV-positive (51.0%) compared to MSM (29.6%). Both populations reported high proportions of ever testing for HIV (95.0% among MSM; 97.1% among TGW). However, a majority of HIV-negative and newly diagnosed HIV-positive individuals did not meet the definition of regular testing (testing every 3 months). While 66.0% of HIV-positive MSM previously knew their status, only 48.1% of HIV-positive TGW previously knew their status. Additionally, low proportions of those who previously knew their HIV-positive status were on ART (77.1% of MSM and 56.0% of TGW).

Significant associations with HIV included identifying as a transgender woman, being aged 25 or older, any reported STI symptoms, and having a reactive syphilis rapid result. With respect to syphilis, 30.5% of MSM and 43.1% of TGW in the survey tested antibody positive. Those who had tested for HIV in the previous year were less likely to test HIV positive.

Most MSM and TGW reported having more than two male partners in the 12 months, with almost one-third (30.8%) reporting five or more. One-quarter (25.1%) of MSM and 53.9% of TGW received payment for sex during the past 12 months. A majority (89.3%) found condoms to be very/ somewhat affordable and 95.4% found them easy/somewhat easy to access, yet 37.3% of TGW and 29.6% of MSM did not use a condom at their last sex act.

Using the AUDIT-C scale, 31.8% of MSM and 28.4% of TGW could be classified as hazardous drinkers. Conversely, drug use was rarely reported for most drugs, except marijuana with 40.2%

of MSM and 52.9% of TGW reporting use at least once in the six months preceding the survey. No participants injected drugs.

The survey found high levels of stigma, discrimination, and indicators of unmet mental health needs. Many TGW reported being refused police service (43.1%) due to their transgender status. A further 83.3% of TGW reported verbal insults, 46.1% reported physical violence, and 17.6% reported rape in the 12 months preceding the survey. On the Patient Health Questionnaire-4 (PHQ-4), 31.0% of TGW had scores suggesting anxiety and 27.6% suggesting depression. Comparatively, 18.4% of MSM scores suggesting anxiety and 14.7% had scores suggesting depression. One-fifth (21.6%) of TGW attempted self-harm in the last six months.

The 876 Study integrated multiple methods to estimate the population size of MSM and TGW, including literature review, the unique object multiplier, service multipliers, the wisdom of the crowds, and a synthesis of methods using the anchored multiplier. Additionally, the study derived estimates of the number of 'reachable' or 'need to be reached' among MSM and TGW as those likely to disclose their sexual orientation or transgender identity and those with unmet HIV prevention or care needs. The anchored multiplier (variance adjusted) synthesized estimates to arrive at a population size for MSM in Jamaica at 42,375 (95% CI: 28,278 – 58,855). The number of MSM likely to disclose or with unmet need was 34,705 (95% CI: 33,163 - 36,080). The synthesized population size for TGW was 3,841 (95% CI: 3,142 – 4,646), with the number likely to disclose or with unmet need as 2,781 (95% CI 2,401 - 3,111).

Based on the findings, we recommend the comprehensive package of prevention services currently offered to Jamaican MSM be tailored in several ways. First, the prevention package should be adapted for younger MSM given the high prevalence of HIV already present among MSM youth. School-based prevention interventions for young MSM and TGW are recommended, along with review of policies that prevent access to HIV care for youth. Second, the extreme burden of HIV experienced by TGW requires that programs specifically target their prevention and treatment needs. Third, the internet and social media can provide greater access to MSM and TGP to deliver interventions enhancing HIV prevention and treatment. Fourth, the national HIV program may capitalize on RDS-like approaches to recruit key affected populations for prevention and treatment interventions. Finally, BSS should be repeated every three to five years to monitor trends in HIV prevalence and risk behaviors among MSM and TGP in Jamaica with rigorous measures of the HIV care continuum.

List of Acronyms

AIDS	Acquired Immune Deficiency Syndrome
ART	Antiretroviral Treatment
ARV	Antiretrovirals
AUDIT-C	Alcohol Use Disorders Identification Test-C
BSS	Bio-behavioral Surveillance Survey
CAPI	Computer-Assisted Personal Interview
CDC	Centers for Disease Control and Prevention
CI	Confidence Interval
CHR	Committee on Human Research
CITI	Collaborative Institutional Training Initiative
ELISA	Enzyme-Linked Immunosorbent Assay
HIV	Human Immunodeficiency Virus
IRB	Institutional Review Board
MOH	Ministry of Health - Jamaica
PEPFAR	U.S. President's Emergency Plan for AIDS Relief
PSE	Population Size Estimation
QDS™	Questionnaire Development System
RDS	Respondent Driven Sampling
RDS-A	RDS-analysis tool
STI	Sexually Transmitted Infection
TGM	Transgender Men
TGP	Transgender Person
TGW	Transgender Women
UCSF	University of California, San Francisco
VL	Viral Load

Introduction and Background

Jamaica is the largest English-speaking island in the Caribbean. The country's population was estimated to be 2.7 million in 2017 (STATIN 2017), with approximately 34,000 living with HIV (MOH 2017). The HIV epidemic may be characterized as both generalized and concentrated because HIV prevalence is estimated at 1.7% in the adult population and over 5% in at least one key population – men who have sex with men (MSM) (MOH 2017). The Government of Jamaica started providing free antiretroviral treatment (ART) to persons infected with HIV in 2002. There have been great strides in improving HIV diagnosis, treatment, and survival; however, reducing incidence remains a challenge. Meanwhile, data on HIV prevalence among MSM and transgender persons (TGP) remain inadequate.

A 2011 integrated biological and behavioral surveillance survey (BSS) with MSM showed 31.7% prevalence; a 2007 BSS showed 32.9% prevalence (Figueroa et al. 2015; Figueroa et al. 2013). Subgroup analysis of the 2011 study revealed an HIV prevalence of 45% among 95 respondents who identified as female (but did not self-identify as transgender), and 52.9% among 17 respondents who self-identified as transgender women (TGW). Previous BSS among MSM in Jamaica also included follow-up HIV testing in a cohort sub-sample. Forty-nine respondents who tested negative in the 2007 survey were identified and retested in the 2011 survey. Eleven were HIV positive in 2011, projecting an annual incidence of 6% (Figueroa et al. 2015).

Unfortunately, data from the 2011 and 2007 BSS may not be generalizable to the larger MSM or TGW populations because they used non-probability-based sampling methods. Although mode of transmission is included on data collection forms used to report HIV cases to the MOH, this information is often left blank or incomplete to protect the patient from stigma, further limiting data on MSM and TGP in Jamaica. For these reasons, and the need for periodic cross-sectional data to track the epidemic and the national response, a repeat BSS survey was implemented in 2017/18. To enhance representation, the BSS used, for the first time in Jamaica, the respondent driven sampling (RDS) methodology.

Rationale for 2017/18 BSS: “The 876 Study”

The 2017/18 BSS of MSM and TGP was branded as “The 876 Study” to avoid exposing MSM and TGP in the name. The 876 Study had a formative, qualitative phase and a quantitative HIV prevalence and behavioral survey phase. Data from the formative phase highlighted challenges faced by TGW including and beyond HIV risk, such as housing, employment, self-care, health care access, and health care provider competency. Similar challenges were reported among MSM, along with high involvement in transactional sex and high rates of physical and sexual abuse. Both groups noted mental health issues (such as depression, substance use and suicidal ideation), lack of sexual health information, fear of knowing their HIV status and perceived stigma and discrimination. The 876 Study explored these issues to increase our understanding of the context of HIV risk among MSM and transgender persons. The quantitative phase therefore included a comprehensive structured interview coupled with HIV testing and syphilis testing.

The current BSS also sought to measure engagement along the continuum of HIV prevention and care among MSM and TGP to gauge success towards Jamaica's “getting to zero” and UNAIDS 90-90-90 goals. “Getting to zero” envisions zero new HIV infections, zero AIDS-related deaths, and zero discrimination. Use of ART plays a pivotal role in these goals. The UNAIDS

90-90-90 targets are that 90% of all people living with HIV to know their HIV status, 90% of all people with diagnosed HIV infection to receive sustained ART, and 90% of all people receiving ART to achieve viral suppression. Achieving VL suppression contributes to longer, healthier lives for persons living with HIV and reduces the risk of sexual transmission to HIV-negative partners. Achieving VL suppression is a direct result of successful engagement in the continuum of services, including HIV testing, care and effective treatment. Filling gaps in the data available from the HIV surveillance system for estimates of the HIV treatment cascade among MSM and TGP was therefore a priority of the 876 Study.

Rationale for Respondent Driven Sampling Method

A review of secondary data along with findings from the formative phase confirmed that MSM and transgender women are stigmatized and difficult to access in Jamaica. Respondents in the formative assessment explained that cultural perceptions and fear of discrimination and violence cause many MSM and TGP to avoid medical and HIV-related services as well as public health research. Difficulty identifying members of the MSM and TGP communities and concerns about unwanted exposure make many conventional sampling methods (i.e., household or phone based surveys) unfeasible.

Among non-conventional methods to sample hidden populations, time-location sampling (TLS) and respondent driven sampling (RDS) are widely used for BSS worldwide. TLS involves sampling at venues where the target population congregates during randomized times and days of the week. RDS is a chain-referral sampling method that uses peers to refer others to the study. Data from the formative phase supported the theoretical underpinnings of RDS as well as practical considerations that favored peer referral over venue-based recruitment. RDS was therefore selected as most appropriate sampling method for Jamaican MSM and TGP.

Rationale for Population Size Estimation Methods

National and subnational estimates of the size of key populations are required for understanding the magnitude of the HIV problem, program planning, resource allocation, monitoring and evaluation, and tracking the impact of the national response on the HIV epidemic.

The 876 Study integrated multiple methods to estimate the number of MSM and TGW in Jamaica and apportion them to regions. In the absence of a gold standard, size estimate methods for hidden populations are imprecise and prone to biases. The use of multiple methods strengthens confidence in estimates, can provide upper and lower acceptable bounds, and reduces the likelihood that the size estimate is unduly biased by limitations of any single method. Triangulating the findings across multiple methods therefore provides plausible, data-driven national and subnational size estimates.

The methods included in this survey were: literature review, the unique object multiplier, service multipliers, the wisdom of the crowds, and a synthesis of the methods using the anchored multiplier. The anchored multiplier is a Bayesian approach which combines estimates produced by several PSE methods coupled to a prior estimate to arrive at a single consensus estimate and a 95% credible interval (Wesson et al. 2018). Many of the size estimation methods have been used elsewhere with guidelines provided by UNAIDS (WHO, 2010) and examples available in the published literature (Okal et al. 2013; Raymond et al. 2013; Maltiel et al. 2015).

Additionally, the study team explored approaches to deriving estimates of the population sizes of 'reachable' or 'need to be reached' among men who have sex with men and transgender

women by estimating the population likely to disclose their sexual orientation or transgender identity and those with specific unmet need.

Goals and Objectives

The overall goal of the 876 Study was to collect representative behavioral and biological information about MSM and TGP in Jamaica relevant to the HIV epidemic and national response.

The specific objectives were to:

1. Determine the prevalence of HIV and assess the potential associations between HIV prevalence and risk behaviors among MSM and TGP in Jamaica
2. Assess engagement along the continuum of HIV prevention and care among MSM and TGP
3. Estimate the population size of MSM and TGP
4. Assess the psychosocial needs of MSM and TGP
5. Recruit a cohort for future HIV incidence surveillance and engagement in care among MSM and TGP in Jamaica
6. Build local capacity to conduct BSS and population size estimation among key populations in Jamaica
7. Strengthen syphilis reporting, referral, and linkage to treatment services.

Methods and Procedures

The 876 Study was a cross-sectional survey of MSM and TGP in Jamaica with methods and procedures adapted from standard BSS protocols used worldwide. A formative assessment determined RDS as the most feasible, inclusive, and potentially representative sampling method for Jamaican MSM and TGP. RDS is a specialized recruitment technique and sampling method developed to approximate probability-based data of hidden populations through long chains of peer referral. The theoretical foundations of RDS have been detailed previously (Johnston 2013). In brief, to achieve reliable estimates, RDS requires that the entire population is connected within a few degrees of separation, that members know each other to be members of the population, that the population is large enough to approximate sampling with replacement, that people recruit approximately randomly from their social networks, and people can state their network size with consistency and accuracy.

RDS begins with the purposeful selection of “seeds” who are known members of the target population who are instructed to recruit a limited number of other target population members from their social network. If these recruits meet the study eligibility criteria, they are enrolled and instructed to recruit other peers from their social network, and so on. Coupons are used to link recruits to recruiters in the data through the use of unique non-identifying codes (Figure 1).

Identification of Seeds

The formative phase identified a pool of potential initial seeds among participants of in-depth interviews and focus group discussions and through referrals from these participants and other stakeholders. We identified seventeen seeds to represent major subgroups and geographic areas. Selected seeds were diverse with regard to:

- Age
- Education or income level
- Area of residence (e.g., four health regions)
- Self-identity (e.g., gay, straight, bisexual, down-low, transgender man, transgender woman, etc.)

BSS Study Population and Eligibility Criteria

The study population included MSM and TGP who met the following eligibility criteria:

- 16 years of age or older
- Biologically male at birth or transgender male or female by self-identity (note: the survey included male-to-female and female-to-male TGP by self-identity regardless of whom they have sex with)
- Had anal or oral sex with a man in the last year (for non-TGP participants)
- Currently residing in Jamaica
- Able to provide informed consent (i.e., communicated in a mutually understood language, not high or drunk, mentally capable)
- Had not previously participated in the survey

Survey Offices

The study operated sites in three fixed locations (Kingston, Ocho Rios, and Montego Bay) with outreach to Portland and Negril. Sites in Ocho Rios, Montego Bay, and Kingston were located in a non-descript buildings (two in medical practices and one in an administrative office). Data

collection in Negril and Portland were in hotels or villas. To avoid stigma from the public, there were no signs to indicate the activities of the survey locations. Each site had a security presence. Only survey staff, investigators, and recruits with valid coupons were granted access beyond the entry area of survey locations.

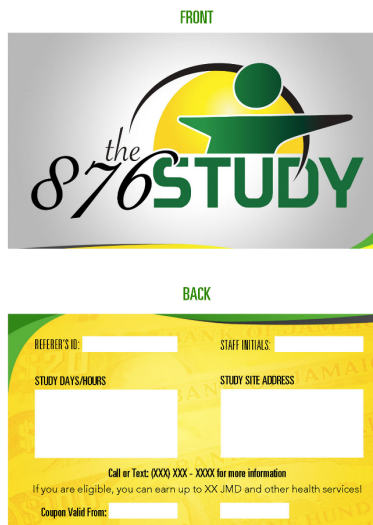
Interview Scheduling

Potential participants received a coupon from their recruiter that provided a study phone number to call or text, drop-in hours of operation, and coupon validity date. Potential participants who called were offered the next available interview slot that was convenient while accommodating drop-ins at the study site. Appointments were scheduled to avoid interaction among participants.

Coupon Management

Upon arrival at the study sites, potential participants were interviewed by the coupon manager who verified coupons and eligibility. The coupon manager created a non-identifying alphanumeric code for each participant based on personal information provided by the participant. The code helped confirm whether the recruit had been enrolled previously. The recruit's eligibility was further assessed through a short personal interview covering the eligibility criteria. A coupon was invalid if tampered with, unreadable, or already used. Invalid coupons were retained and stamped "VOID". Valid coupons of potential participants undergoing screening for eligibility were retained and stamped "USED".

Figure 1 Sample RDS Coupon



Following completion of study procedures, participants met with the Coupon Manager who explained the recruitment process and the handling of the peer recruitment coupons. Participants were carefully trained to only share the coupon number and the contact number for the study with persons they knew personally and not to give the coupon to more than one person.

Participants also had the option of sharing coupon information electronically via email or chat app (WhatsApp, etc.) once they have introduced the study to persons in their network. Findings from the formative phase supported the use of electronic coupons along with paper. Similar

adaptations have been used for RDS surveys among MSM in Kampala, Uganda and San Francisco, USA (Truong et al., 2013; Kajubi et al. 2013).

Initially, participants were given three to five coupons each. During slow recruitment periods, the number of coupons were increased up to 20. Once the sample size approached the target and equilibrium was achieved on key variables, coupon dispersal to remaining participants was phased out.

Issuance and receipt of coupons were monitored electronically using RDS Coupon Manager software. Codes were pre-printed on coupons and used to track participants, their recruits, and site of origin. The linkage between enrolled participants and those they referred to the study was preserved for statistical analysis. The same coupon code was used to link behavioral data from the questionnaire to test results.

Informed Consent

After eligibility screening, potential participants received a printed copy of the informed consent form, which the study interviewer verbally reviewed to facilitate understanding. The 876 Study procedures, risks, benefits, and contact information were explained. Persons interested in participating were asked to provide informed consent by signing their names. Separate sections recorded consent for: 1) the risk behavior questionnaire with rapid testing for HIV and syphilis, 2) dried blood spot (DBS) card storage for up to five years for future HIV and STI tests also stipulating that the participant can have their specimen removed and destroyed at any time, and 3) participation in follow-up activities by the MOH. The minimum level of participation for the 876 Study included completion of the questionnaire and rapid HIV and syphilis testing. Individuals were informed that they could terminate or withdraw their participation in any or all of the 876 Study activities on the current day and, should they consent to recontact, at any time during the follow-up period.

Survey Data Collection

The behavioral questionnaire was interviewer-administered on tablet PC units. Standardized data collection instruments were adapted for MSM and TGP in Jamaica. These instruments collected data on demographic characteristics, behaviors potentially correlated with HIV, symptoms of STI, awareness use of prevention and other services, prior HIV testing and engagement in the HIV continuum of care, exposure to interventions, as well as HIV-related knowledge, attitude, practices, stigma, discrimination, and perceptions. Data items also included indicators to track the HIV epidemic and the national response among MSM and TGP, conforming to international standards (e.g., UNAIDS GAM indicators, local Key Performance/M&E Indicators), UNAIDS 90-90-90 targets, national program needs, previous Jamaican MSM surveys, and similar surveys in the region.

Pre-test/Risk Reduction Counseling

Upon completion of the questionnaire and prior to collection of specimens for testing, participants received pre-test/risk reduction counseling for HIV and syphilis based on Jamaica national guidelines. Pre-test/risk reduction counseling included sensitization to issues affecting MSM and TGP, explanation of HIV infection and transmission, the meaning of test results, risks associated sexual behaviors, and means to prevent and treat HIV and STI.

Biological specimens for HIV and syphilis screening were collected according to Jamaican MOH protocols. Participants with reactive rapid tests were referred for follow-up visits with private physicians or at public treatment sites. In keeping with the Public Health Act, trained testers

completed Class 1 notification forms for persons who were reactive for HIV or syphilis. The site coordinator delivered completed forms to the Medical Officer of Health at the Parish Health Department on a weekly basis.

Rapid HIV Testing

Rapid HIV testing was conducted using finger-prick whole blood at the survey site. Eligible and consenting participants were tested using Determine rapid test kits (Alere Determine™ HIV-1/2 rapid test kits), Unigold (Uni-Gold™ HIV, Trinity Biotech PLC, Ireland), and Oraquick, with Determine as the first test, Unigold as the second test, and Colloidal Gold as the third test. Participants who were non-reactive on Determine were recorded as negative. Participants with specimens that were reactive on Determine were tested again using Unigold. Persons who were reactive both (Determine+, UniGold+), were recorded as HIV positive. For persons who were reactive on Determine but non-reactive on Unigold (Determine +, Unigold-) the result was recorded as inconclusive and a third Oraquick test was administered. If the third assay was reactive (Determine +, Unigold-, Oraquick +), an HIV-inconclusive result was recorded, the participant was referred to a treatment site/private practitioner to be retested in 14 days with followed up by study staff. If the third assay was non-reactive (Determine +, Unigold-, Oraquick -) their HIV status was recorded as negative. The site coordinator maintained a list of participants with 'inconclusive' HIV test results. The coordinator followed-up with the contact person at each referral site to obtain the results of confirmatory tests and update study records.

Syphilis Testing

Rapid syphilis testing was conducted using whole blood collected by finger prick. Samples were collected following standard operating procedures of the MOH for outreach testing. On-site testing used SD Bioline Syphilis 3.0. If the SD Bioline test was reactive, participants were told the test result, counseled, and given referral to conduct confirmatory testing at the nearest Public Health Facility or a private clinician of their choice. If the test was non-reactive, the participant was told their result and counseled on risk reduction.

Dried Blood Spot (DBS) Storage for Additional and Future Testing

Participants could also consent or refuse to provide DBS specimen collection which would be stored for future tests. Consenting participants provided five spots of ~70-75 µl of blood on a DBS card (Whatman 903 paper [Whatman, Inc., Piscataway, NJ]) collected via fingerstick. DBS cards were labeled with the unique testing code (UTC), target group (TGM, TGW, MSM), and age, date sample was prepared, coded HIV status, self-reported ARV treatment status (coded for treatment naive, previous treatment, current treatment).

DBS Cards were dried at ambient temperature overnight. After drying, each DBS card was individually wrapped in a folded sheet of glassine paper. Five to ten glassine paper-wrapped cards were packaged into a low-gas-permeable, zip-closure plastic bag containing eight to ten desiccant bags, one humidity indicator (Fisher Scientific, Inc., Pittsburgh, PA), and specimen inventory form. The packaged DBS cards were kept in a dry and safe location at ambient temperature (18-27°C) with no direct exposure to sunlight until shipment to the NPHL. The packages of DBS cards were delivered to the NPHL on a weekly basis. At the end of the mobile data collection, the packaged DBS cards were transported back to the fixed site where they were stored until they are delivered to the NPHL on the next scheduled weekly delivery date. The DBS cards were frozen at -70°C or below for long-term storage at the NPHL.

Participants are able to withdraw their specimens from storage to be permanently destroyed at any time by contacting the Senior Medical Officer, HIV/STI/TB at the MOH. Specimens will be retained for five years after the conclusion of the study, after which time they will be destroyed.

Viral Load Results

The 876 Study linked viral load results by requesting data from the national Monitoring & Evaluation (M&E) unit at the MOH for HIV-positive participants who provided written consent. At the end of data collection, a list including participant's name, reported current treatment site, and month and year of birth was submitted to the MOH to identify participants' VL results in the treatment site database. Viral load results were encrypted and shared in password protected files with the study site coordinator. The site coordinator provided the database manager with a list of VL results, organized by UTC for merging with the survey data. This ensured names were not linked to behavioral questionnaires or HIV and syphilis results.

Post-test Counseling and Linkage to Services

Results disclosure, post-test counseling, and referral to care and treatment were provided by the counselor immediately following rapid testing. Counseling messages followed Jamaica national HIV guidelines and were tailored to participants' results and risk profiles. Post-test counseling included goals, means, and strategies for behavioral risk reduction (e.g., condom use, reducing multiple concurrent partners, etc.).

Referrals

The study referred participants to the nearest Public Health Facility or a private clinician of their choice using procedures detailed in the national Referral and Linkage Protocol. Study sites had a list of LGBT-friendly providers for participants who did not wish to receive care at a public facility. The site coordinator completed a referral form, which the participant was instructed to bring to the identified point person (usually the contact investigator or social worker) at the referral facility. The site coordinator would also assist in scheduling appointments, giving directions to facilities, providing a transportation voucher if needed, sending contact details of the participant to the referral facility, and following-up to ensure contact. Participants were also offered support from a peer navigator. Peer navigators are trained and employed by partner agencies to link persons with positive or reactive test results in outreach settings to treatment facilities. The Referral and Linkage Protocol instructs that in the event that a newly diagnosed patient does not show up for the first doctor's visit, the staff at the referral facility should track the patient using the LTFU protocol.

Incentives

For the primary incentive (i.e., for their own study enrollment), participants received JMD\$1500 (~\$12.50 USD) and up to JMD\$1000 travel reimbursement (~\$8.30 USD) for travel from up to two parishes away or for commute by private taxi. Participants received secondary incentives (i.e., for each successful enrollment of an eligible recruit) of cell phone airtime of JMD\$200 (~\$1.60 USD). Participants also received a pouch of non-monetary incentives that included condoms, lubricants, informational materials, and items for personal hygiene (i.e., bath soap, washrags, deodorant). Finally, participants were entered into a raffle with a draw for one winner for every 30 participants. Raffle prizes included event tickets and gift certificates valued at ~\$50 USD. Winners selected their preferred prize from available options.

Staff Training

Study staff were trained and certified for HIV counseling and testing with specific training in DBS collection. Staff involved with handling and analyzing data were trained in collection, management, security, and analysis procedures.

Survey Sample Size Calculation

This study set a target sample size of 646 MSM and TGP based on an acceptable margin of error ($\pm 5\%$) on the estimated HIV prevalence of 30% (i.e., observed in the previous BSS) with a design effect of 2 using the following formula:

$$N = \frac{(t^2 * p(1 - p))}{m^2} * DE$$

where “N” is the required sample size, “t” is the confidence interval level, “p” is estimated prevalence, “m” is the margin of error, and DE is the design effect.

Data Management and Storage

Error checks and skip patterns were programmed into the electronic questionnaires, including completeness, consistency checks, and detection of illogical values. The electronic-questionnaires were pre-tested to ensure that skip patterns were correct. Continuous quality checks were performed to ensure that code numbers were recorded properly. At the end of each interview, the interviewer uploaded the file from the tablet to the study server. After a successful upload, the questionnaire was deleted from the tablet.

Study files were continuously synced to a secured and encrypted cloud storage folder. Files included the coupon logs, HIV and syphilis testing log, and site coordinator log. These files were backed up to an encrypted hard drive on a daily basis by the data manager. On a weekly basis, interviews were downloaded from the study server and merged with the other logs. Data quality checks were done on the merged data to ensure consistency and accuracy. The merged dataset was loaded into Stata (v14) where summary statistics were obtained and used for weekly report generation using an Excel template.

Analysis

As a surveillance effort, primary analyses are point estimates of disease prevalence and key risk behaviors (e.g., HIV, STI, unprotected sex), and use and access to prevention and care programs. Analyses include crude estimates (i.e., unadjusted sample means and proportions) and estimates that account for the RDS sampling design. Results are presented for the sample as a whole (that is, MSM + TGP) and separately for MSM and TGW, given major differences in HIV prevalence and key indicator variables for these groups. The sample size for TGM, however, was not sufficient for separate analysis.

Analyses were conducted in RDS-Analyst (RDS-A) for adjusted point estimates and confidence intervals (available from http://wiki.stat.ucla.edu/hpmrg/index.php/RDS_Analyst_Install, SAS, Version 9.2, Cary, NC). Using exported RDS weights, multivariate logistic regression was done to identify significant correlates of HIV infection, in Stata (Version 10, College Station, TX). Given that all three populations were closely networked (i.e., MSM+TGW+TGM), RDS-adjustments were limited to the entire sample.

A few errors detected in the data are noted. Four participants who tested HIV-negative but self-reported being HIV-positive and who were not found in MOH registers were dropped from analysis due to uncertainty as to their status. One participant who tested HIV-negative, self-

reported being HIV-positive and was found engaged in care from MOH registers was included as HIV-positive. The initial survey specified that PrEP questions should only be asked of self-identified HIV-positive individuals, which was only later identified as incorrect (i.e., the question was intended for self-identified HIV-negative respondents. The error was rectified upon realization, but resulted in a high number of 'missing' cases for the early part of the survey.

For construction of the HIV care cascade we used two sources of data. In the questionnaire, participants were asked about their last HIV test result, whereby we gleaned information on those who already knew they were positive. Participants who knew they were positive were asked about current ART use. For viral load, a list of named participants who had previously consented to being included was sent to MOH. The list of 591 records resulted in 106 people matched to an existing clinic record. Participants who were found on ART in clinic records but who had not disclosed HIV status in the interview were recoded as previously known positive and on ART. In the clinic records, dates of last viral load test were incomplete. As a result, some participants may have a matched VL after they were in the study. As such, viral load estimates should be seen as slightly overestimated. To rectify, CDC Atlanta will retest DBS samples at headquarters and results will be updated.

Ethical Considerations

The protocol was reviewed and approved by the Ethical Review Boards of the Jamaica MOH and UCSF, and by CDC's Scientific Integrity Branch in the Division of Global HIV & TB. Study staff completed CITI ethical training.

The following steps were taken to mitigate risks to participants:

- Staff were professionally trained and certified as HIV counselors, under close supervision.
- Staff were trained on specific study measures to keep data confidential and secure.
- Names or other identifying information were not linked to survey responses.
- Survey data were stored in password-protected files and tablets and computers were encrypted and password protected.
- Paper-based study materials were stored in locked file cabinets, in locked offices.
- Staff did not ask for identification (such as government issued ID) from any participant.
- Staff working with participants were required to sign an Employee Confidentiality Agreement.

RESULTS

RDS Recruitment Outcomes

An initial seven seeds were identified across the three sites to launch recruitment (Table 1). As the study progressed, ten additional seeds were added, selected for diversity in gender, socio-economic status, and parish of residence. Over the 285 days of recruitment, a maximum of 16 waves were completed. A majority of the sample participated in the study at the Kingston site office. The sample included 537 self-identified MSM, 102 TGW, and 13 TGM. Results in the text of the present report describe measures in the sample (i.e., RDS-A unadjusted) to provide key indicators for MSM and TGW participants. RDS-A results are presented for the theoretical combined population as a whole given the high level of inter-group recruitment which is needed for calculations of adjustments for point estimates and confidence intervals.

Table 1: RDS recruitment

Site	Days to Recruit	Number of Seeds	Number of Waves	Sample Size
Kingston	285	7	16	360
Ocho Rios	220	6	9	108
Montego Bay	202	4	16	184
Total	285	17	16	652

Figure 2 illustrates the recruitment conforms to RDS theoretical requirements in that chains progressed to multiple waves, achieving independence from the selected seeds, with cross-recruitments between MSM and transgender persons (i.e., they form an interconnected social network). Equilibrium (i.e., stability in the composition of the sample) was also achieved on key variables tracked during the course of the studying, including demographic characteristics, parish of residence, HIV serostatus, syphilis, prior HIV testing history, and access to prevention programs. No bottlenecks (i.e., substantive differences between different chains) were noted.

Figure 2 Recruitment diagram by sex/gender



Demographic Characteristics

A majority of the sample were born male (98.6%) (Table 2, crude or sample estimates are cited in text, population-adjusted estimates are in Tables unless otherwise specified). However, 12.8% of MSM reported a current gender other than male, with 11.7% identifying as gender queer. All transgender women identified as male-to-female as current gender. Over half the sample (54.6%) was under the age of 25 years, with the mean age of MSM being 25.7 and for TGW 24.3 years. The youngest participant was 16 years (youngest allowed per protocol) and the oldest was 65 years. More than half (51.6%) of MSM identified as bisexual, with 46.7% identifying as gay. TGW largely identified as being gay or lesbian (77.5%), 9.8% identified as heterosexual/straight. Most participants completed at least some or graduated from secondary school (98.1%), with 11.8% having university or higher education and 24.8% having vocational training. Some participants were full-time (16.4%) or part-time (6.0%) students.

Table 2: Demographics of MSM and TGP

		MSM (n=537)		TGW (n=102)		Total (n=652)			
		Crude		Crude		Crude		RDS-Adjusted	
		n	%	n	%	n	%	%	95% CI
Sex at Birth									
	Male	536	99.8	102	100	643	98.6	99.0	97.9->99.9
	Female	1	0.2	0	0	9	1.4	1.0	<.01-2.1
Current Gender									
	Male	468	87.2	0	0	468	71.8	75.5	69.7-81.5
	Female-to-male	0	0	0	0	13	2.0	1.1	0.2-2.0
	Male-to-female	0	0	102	100	102	15.6	11.5	7.3-15.6
	Genderqueer	63	11.7	0	0	63	9.7	11.2	6.7-15.7
	Other	6	1.1	0	0	6	0.9	0.6	<.01-1.6
Age									
	16-17	29	5.4	5	4.9	34	5.2	6.6	2.4-10.8
	18-24	266	49.5	51	50.0	322	49.4	52.4	45.6-59.3
	25+	242	45.1	46	45.1	296	45.4	41.0	34.0-47.9
Age									
	16-17	29	5.4	5	4.9	34	5.2	6.6	2.4-10.8
	18-24	266	49.5	51	50.0	322	49.4	52.4	45.5-59.3
	25-29	129	24.0	30	29.4	161	24.7	22.3	17.2-27.5
	30-39	75	14.0	16	15.7	95	14.6	11.4	7.5-15.3
	40-49	29	5.4	0	0	31	4.8	6.1	2.5-9.8
	50+	9	1.7	0	0	9	1.4	1.2	<.01-2.4
Sexual Orientation									
	Gay/Lesbian	251	46.7	79	77.5	338	51.8	46.8	40.2-53.5
	Bisexual	277	51.6	13	12.7	295	45.2	50.6	43.9-57.2
	Heterosexual/Straight	8	1.5	10	9.8	18	2.8	2.6	<.01-5.3
	Missing	1	0.2	0	0	1	0.2	--	--
Highest education level completed									
	Primary or less	8	1.5	2	2.0	10	1.5	1.5	<.01-3.3
	Any secondary	334	62.2	58	56.9	401	61.5	59.1	52.3-65.9
	University or higher	68	12.7	8	7.8	77	11.8	10.5	7.0-14.1
	Vocational	125	23.3	34	33.3	162	24.8	28.4	22.2-34.5
	Other	2	0.4	0	0	2	0.3	0.5	<.01-1.3
Current student status									
	Full-time student	92	17.1	14	13.7	107	16.4	16.5	11.3-21.7
	Part-time student	31	5.8	7	6.9	39	6.0	5.8	3.4-8.2
	Not a student	414	77.1	81	79.4	506	77.6	77.7	72.0-83.4

A plurality of the sample (40.2%) had a monthly household income of less than 40,000 JMD (<\$300 USD), (46.1% for TGW and 38.9% for MSM) (Table 3). An indication of food insecurity (skipping or reduced meals due to lack of money) was found in 29.4% of MSM and 33.6% of TGW. Unemployment was also high at 38.5% (47.1% for TGW and 37.1 for MSM). The largest sector of employment for MSM was professional (21.0%), which comprise bankers, government workers, sales representatives, and similar occupations. An additional 14.5% of MSM reported working in services industries, including hotel workers, retail staff, and performers. For TGW, 12.7% reported working in service industries, which constituted their largest sector of employment. Sex work was reported by 9.8% of TGW as their main occupation.

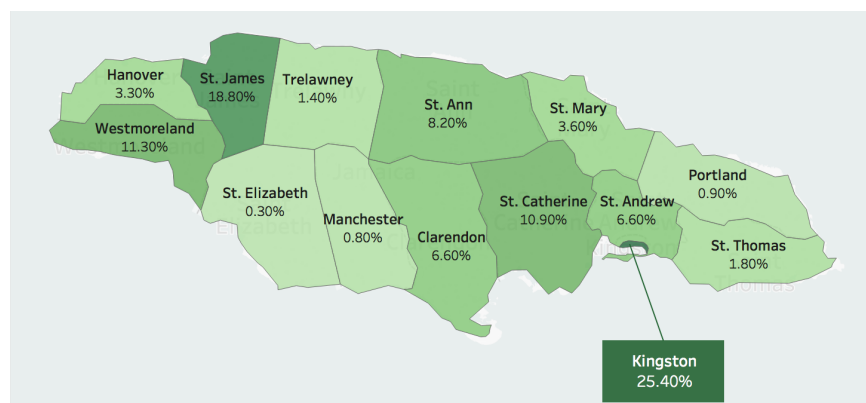
Table 3: Income, Occupation, and Parish of Residence of MSM and TGP

		MSM (n=537)		TGW (n=102)		Total (n=652)			
		Crude		Crude		Crude		RDS-Adjusted	
		n	%	n	%	n	%	%	95% CI
Household Monthly Income									
	<=40,000 JMD	209	38.9	47	46.1	262	40.2	48.2	42.4-53.89
	41k - 149k JMD	169	31.5	24	23.5	195	29.9	35.9	31.0-40.8
	150k - 299k JMD	48	8.9	4	3.9	52	8.0	9.3	6.5-12.2
	300k - 499k	16	3.0	2	2.0	20	3.1	4.0	2.1-5.8
	>=450k	11	2.0	2	2.0	13	2.0	2.6	1.1-4.1
	Do not know	67	12.5	18	17.6	87	13.3	--	--
	Declined	15	2.8	5	4.9	21	3.2	--	--
Food insecurity: Skipped or reduced meals due to lack of money									
	Yes	158	29.4	39	33.6	197	30.2	27.2	21.3-33.2
	No	379	70.6	77	66.4	456	69.9	72.8	66.8-78.3
Occupation									
	Not employed	199	37.1	48	47.1	251	38.5	41.6	37.7-45.4
	Professional	113	21.0	8	7.8	123	18.9	17.2	14.5-19.8
	Laborer	41	7.6	8	7.8	50	7.7	7.6	4.9-10.3
	Agriculture/fishing	5	0.9	3	2.9	9	1.4	1.2	0.2-2.2
	Driver	7	1.3	2	2.0	9	1.4	1.2	0.3-2.2
	Retail/restaurant/performer	78	14.5	13	12.7	95	14.6	14.0	10.6-17.3
	Security/military/police	28	5.2	3	2.9	31	4.8	4.4	2.0-6.8
	Sex work	12	2.2	10	9.8	22	3.4	7.8	4.6-11.0
	Call center	43	8.0	3	2.9	47	7.2	7.8	4.6-11.0
	Other	10	1.9	4	3.9	14	2.1	1.8	<.01-4.7
	Missing	1	0.2	0	0	1	0.2	--	--
Parish of Residence									
	Clarendon	36	6.7	16	15.7	52	8.0	6.6	<.01-14.2
	Hanover	13	2.4	1	1.0	14	2.1	3.3	<.01-6.8
	Manchester	3	0.6	2	2.0	5	0.8	0.8	<.01-1.8
	Portland	5	0.9	0	0	6	0.9	0.9	<.01-3.3
	Kingston & St. Andrew/	200	37.2	41	40.2	248	38.0	38.3	19.8-56.9

St. Ann	33	6.1	15	14.7	48	7.4	8.2	0.5-15.9
St. Catherine	71	13.2	13	12.7	88	13.5	10.9	3.6-18.2
St. Elizabeth	3	0.6	0	0	3	0.5	0.3	<.01-0.9
St. James	103	19.2	3	2.9	106	16.3	18.8	3.2-34.5
St. Mary	9	1.7	3	2.9	12	1.8	3.6	<.01-7.5
St. Thomas	13	2.4	0	0	14	2.1	1.8	<.01-4.3
Trelawney	4	0.7	4	3.9	8	1.2	1.4	<.01-3.4
Westmoreland	44	8.2	4	3.9	48	7.4	11.3	1.5-21.2

The study recruited participants from every parish in Jamaica (Table 3). The largest proportion (38.0%) of MSM and TGP lived in Kingston & St Andrew, followed by St James (18.8%). A majority of MSM and TGP resided in and around the urban areas of Kingston, Ocho Rios, Montego Bay, and Negril. Figure 3 shows the RDS-adjusted parish of residence reflecting the distribution of MSM in Jamaica.

Figure 3: RDS-Adjusted Parish of Residence



Sexual Behavior

Most MSM and TGP reported having multiple male partners in the 12 months preceding the survey, with 30.0% of MSM and 37.3% of TGP having 5 or more male partners (Table 4). Nonetheless, 22.5% of MSM and 23.5% of TGP reported having only one male partner. Having at least one female partner in the last 12 months was reported by 38.5% of MSM and 17.6% of TGP. Self-reported sex work, defined as having received cash payment for sex in the previous 12 months, was reported by 25.1% of MSM and 53.9% of TGP. Additionally, 12.1% of MSM and 19.6% of TGP reported having paid for sex in the previous 12 months. The preferred sexual practice was versatile anal intercourse (insertive and receptive) for MSM (47.9%) and receptive anal intercourse for TGP (50.0%). A substantial proportion of TGP nonetheless preferred versatile (40.2%) or insertive anal sex (9.8%). A majority of both MSM (62.9%) and TGP (59.8%) had a sexually debut by age 15 years, with the mean age of debut being 14.4 years for both.

Table 4: Sexual Behavior of MSM and TGW									
		MSM (n=537)		TGW (n=102)		Total (n=652)			
		Crude		Crude		Crude		RDS-Adjusted	
		n	%	n	%	n	%	%	95% CI
Total number of male partners in previous 12 months									
	0	8	1.5	0	0	13	2.0	2.5%	<.01-5.3
	1	121	22.5	24	23.5	148	22.7	22.9	17.4-28.5
	2	118	22.0	16	15.7	136	20.9	23.3	17.8-28.5
	3	88	16.4	12	11.8	100	15.3	17.6	12.2-23.1
	4	41	7.6	12	11.8	54	8.3	9.2	5.1-13.3
	5 or more	161	30.0	38	37.3	201	30.8	24.5	24.5-29.6
Total number of female partners in previous 12 months									
	0	330	61.5	84	82.4	416	63.8	60.4	53.4-67.2
	1	69	12.8	7	6.9	79	12.1	10.0	6.6-13.3
	2	41	7.6	2	2.0	45	6.9	9.0	4.6-13.5
	3	30	5.6	4	3.9	35	5.4	7.7	3.4-11.9
	4	12	2.2	0	0	12	1.8	1.6	0.3-2.9
	5 or more	53	9.9	5	4.9	63	9.7	11.3	6.3-16.4
Paid for sex with any partners in previous 12 months									
	Yes	65	12.1	20	19.6	87	13.3	12.6	7.9-17.2
	No	472	87.9	82	80.4	565	86.7	87.4	82.8-92.1
Received payment for sex with any partners in previous 12 months									
	Yes	135	25.1	55	53.9	195	29.9	24.3	18.7-30.0
	No	402	74.9	47	46.1	457	70.1	75.7	70.0-81.3
Sexual position preference									
	Receptive	135	25.1	51	50.0	188	28.8	29.0	22.5-35.5
	Insertive	135	25.1	10	9.8	145	22.2	26.4	21.5-31.2
	Oral Only	8	1.5	0	0	12	1.8	2.3	<.01-5.3
	Versatile	257	47.9	41	40.2	304	46.6	41.9	35.5-48.3
	None	1	0.2	0	0	2	0.3	0.4	<.01-1.1
	Missing	1	0.2	0	0	1	0.2		
Age at sexual debut									
	≤10 years	69	12.8	12	11.8	83	12.7	14.5	9.6-19.5
	11-15	269	50.1	49	48.0	322	49.4	50.5	44.0-56.9
	16-17	111	20.7	26	25.5	138	21.2	19.4	14.1-24.6
	18+	82	15.3	15	14.7	98	15.0	15.6	11.0-20.3
	Missing	6	1.1	0	0	11	1.7		

Venues where MSM and TGW find sexual partners are shown in Figure 4. For both, the internet was the number one place to find sexual partners, followed by friends for MSM and the street for TGW. Parties and limes were also common places to find sexual partners for MSM and TGW.

Figure 4: Venues for Meeting Sexual Partners

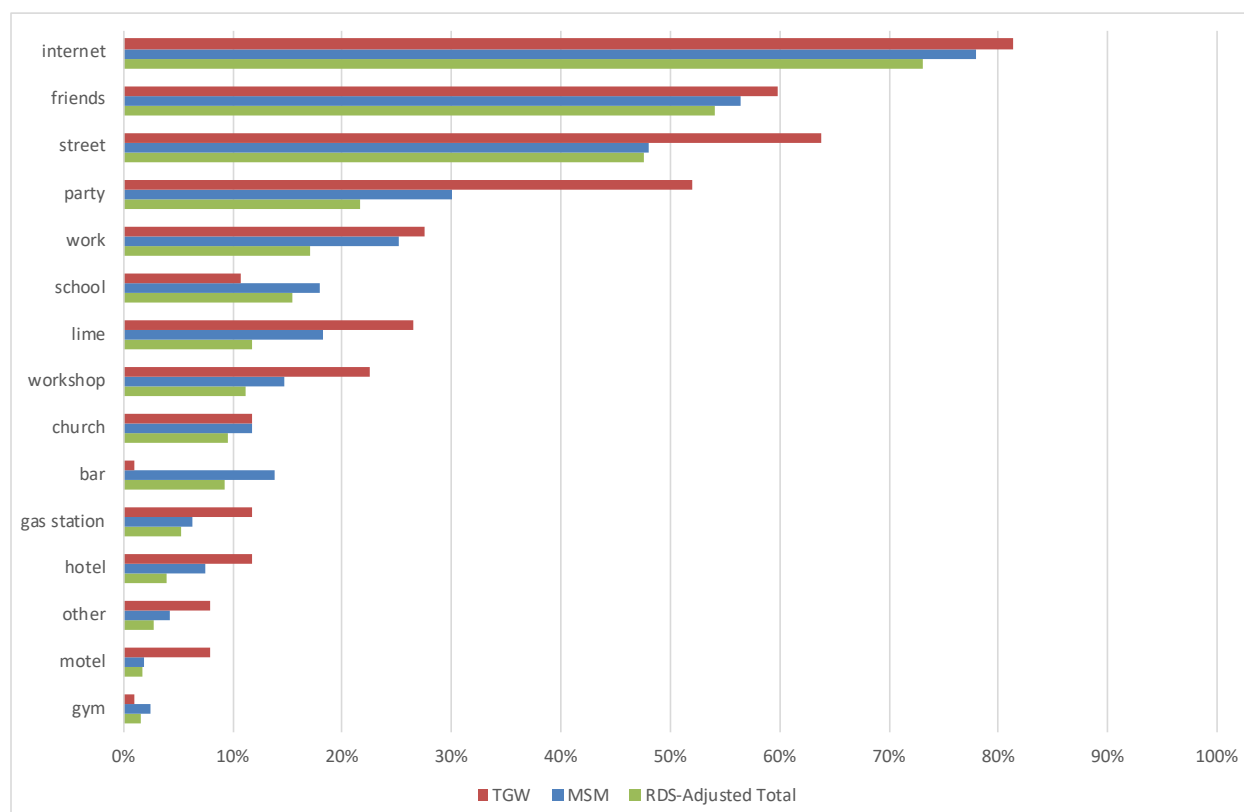


Table 5 shows indicators of access, potential barriers, and use of condoms. Condoms not being affordable or only somewhat affordable was reported by 29.1% of MSM and 42.2% of TGW in the survey. Condoms not being easily accessible or only somewhat easy to access was reported by 32.1% of MSM and 39.3% of TGW. Majorities of MSM (73.4%) and TGW (79.4%) had received a free condom. More TGW reported ever using a female condom than MSM (9.8% vs 2.0%, respectively). Condom use at last sex was reported by 65.4% of MSM and 57.8% of TGW.

Table 5: Condom Affordability, Accessibility, and Lubricant Use

		MSM (n=537)		TGW (n=102)		Total (n=652)			
		Crude		Crude		Crude		RDS-Adjusted	
		n	%	n	%	n	%	%	95% CI
Condom Affordability									
	Very affordable	366	68.2	55	53.9	426	65.3	65.3	61.0-69.6
	Somewhat affordable	117	21.8	32	31.4	156	23.9	23.9	20.1-27.7
	Not affordable	39	7.3	11	10.8	51	7.8	7.5	5.2-9.7
	Don't Know	15	2.8	4	3.9	19	2.9	3.4	1.5-5.3
Condom Accessibility									
	Easy to access	360	67.0	59	57.8	424	65.0	64.8	60.2-69.4
	Somewhat easy to access	157	29.2	33	32.4	198	30.4	31.0	26.6-35.5
	Not easy to access	15	2.8	7	6.9	22	3.4	2.8	1.5-4.0
	Do not know	4	0.7	3	2.9	7	1.1	1.1	0.2-2.1
	Missing	1	0.2	0	0	1	0.2		
Received Condom for Free									
	Yes	394	73.4	81	79.4	482	73.9	75.1	71.3-78.8
	No	142	26.4	21	20.6	169	25.9	24.8	21.1-28.5
	Don't know	1	0.2	0	0	1	0.2		
Ever Used Female Condom									
	Used female condom	11	2.0	10	9.8	23	3.5	3.7	2.2-5.2
	Did not use	525	97.8	92	90.2	628	96.3	96.2	94.7-97.7
	Missing	1	0.2	0	0	1	0.2		
Condom Use* at Last Sex Act									
	Yes	351	65.4	59	57.8	416	63.8	66.4	62.5-70.4
	No	159	29.6	38	37.3	203	31.1	33.6	29.6-37.5
	Missing	27	5.0	5	4.9	33	5.1		
Frequency of Lubricant Use									
	Always	190	35.4	42	41.2	236	36.2	35.5	31.3-39.7
	Usually	93	17.3	12	11.8	106	16.3	17.5	14.2-20.9
	Sometimes	160	29.8	32	31.4	195	29.9	29.8	26.1-33.6
	Rarely	50	9.3	13	12.7	65	10.0	10.0	7.4-12.5
	Never	44	8.2	3	2.9	50	7.7	7.2	4.9-9.5

*For TGP who reported a TGP partner, they were asked about 'method of prevention (dental dam or condom)' rather than only condom use but are included here.

Alcohol and Drug Use

Table 6 shows two measures of potential alcohol misuse. Having six or more drinks at one sitting, an indicator of binge drinking, was reported once a month or more by 19.3% of MSM and 17.6% of TGW. Audit-C scores indicate hazardous drinking among 31.8% of MSM and 28.4% of TGW. Apart from marijuana, with at least one use reported by 40.2% of MSM and 52.9% of TGW, use of most drugs was infrequently acknowledged and are not shown. No MSM or TGW said they had ever injected drugs.

Table 6: Drug and Alcohol Use among MSM and TGP									
		MSM (n=537)		TGW (n=102)		Total (n=652)			
		Crude		Crude		Crude		RDS-Adjusted	
		n	%	n	%	n	%	%	95% CI
How often has 6 or more drinks									
	Never	315	58.7	66	64.7	388	59.5	60.4	56.1-64.6
	Less than once a month	116	21.6	17	16.7	135	20.7	20.1	16.1-24.1
	Monthly	49	9.1	4	3.9	56	8.6	8.7	6.2-11.1
	Weekly	43	8.0	6	5.9	50	7.7	7.4	3.0-11.7
	Daily	12	2.2	8	7.8	21	3.2	3.5	1.5-5.5
	Missing	1	0.2	1	1.0	2	0.3		
Audit-C Designation									
	Hazardous drinker	171	31.8	29	28.4	204	31.3	30.7	26.7-34.8
	Not a hazardous drinker	365	68.0	73	71.6	448	68.7	69.3	65.2-73.3
Frequency of marijuana use									
	Did not use	320	59.6	48	47.1	372	57.1	57.5	54.5-60.6
	Used at least once	216	40.2	54	52.9	279	42.8	42.5	39.4-45.5
	Declined/missing	1	0.2	0	0	1	0.2		

STI Diagnosis, Symptoms, and Treatment

Self-reported STI symptoms and treatment were common (Table 7). Among MSM, 25.7% said they were diagnosed with an STI (regardless of whether they had symptoms) and 21.6% reported symptoms of STI (regardless of whether they were treated). Among TGW, 30.4% said they were diagnosed and 30.4% had symptoms of STI (note: 32 TGW had either). Among MSM who were either diagnosed or had symptoms, 85.6% received treatment. The comparable treatment figure for TGW was 72.7%.

Table 7: STI Diagnosis, Symptoms, and Treatment									
		MSM (n=537)		TGW (n=102)		Total (n=652)			
		Crude		Crude		Crude		RDS-Adjusted	
		n	%	n	%	n	%	%	95% CI
Clinician diagnosed STI in previous 12 months									
	Yes	138	25.7	31	30.4	170	26.1	20.4	15.4-25.4
	No	398	74.1	70	68.6	479	73.5	79.4	74.4-84.4
	Don't know/missing	1	0.2	1	1.0	3	0.5		

STI symptoms in previous 12 months									
	Yes	116	21.6	31	30.4	148	22.7	19.5	12.7-26.3
	No	417	77.7	70	68.6	498	76.4	80.3	73.5-87.1
	Don't know/missing	1	0.2	1	1.0	3	0.5	0.2	<.01-0.7
	Missing	3	0.6	0	0	3	0.5		
Treated for STI (out of those with symptoms or diagnosis)									
	Yes	149	85.6	32	72.7	182	83.1	84.3	78.9-89.7
	No	25	14.4	12	27.3	37	16.9	15.7	10.3-21.1

Stigma, Discrimination, and Incarceration

Stigma and discrimination manifested as refusal of services and verbal, physical, and sexual violence due to MSM and TGW status, with multiple events common even within the last 12 months (Table 8). These events were particularly frequent among TGW. The most common refusal of service for both populations was by the police (15.8% for MSM and 43.1% for TGW), followed by discrimination in housing (12.3% for MSM and 31.4% for TGW). Majorities of TGW (83.3%) and MSM (64.6%) experienced verbal abuse in the last 12 months. Additionally, almost half (46.1%) of TGW experienced physical violence, and 17.6% reporting sexual violence. Among MSM, 29.1% had ever been to jail or prison, while 42.2% of TGW reported the same.

Table 8: Stigma, Discrimination, and Incarceration among MSM and TGP									
		MSM (n=537)		TGW (n=102)		Total (n=652)			
		Crude		Crude		Crude		RDS-Adjusted	
		n	%	n	%	N	%	%	95% CI
Refused services in previous 12 months due to MSM/transgender status									
	Healthcare	12	2.2	10	9.8	22	3.4	2.5	1.0-3.9
	Employment	46	8.6	28	27.5	74	11.3	7.3	4.8-9.9
	Religious services	39	7.3	23	22.5	62	9.5	6.9	4.1-9.7
	Service (i.e. restaurant)	29	5.4	28	27.5	57	8.7	4.4	2.4-6.4
	Housing	66	12.3	32	31.4	98	15.0	12.6	8.3-16.8
	Police assistance	85	15.8	44	43.1	129	19.8	14.8	10.0-19.7
Experienced verbal insults in previous 12 months									
	Yes	347	64.6	85	83.3	432	66.3	61.9	55.0-69.0
	No	188	35.0	17	16.7	205	31.4	37.8	30.7-44.7
	Don't know/missing	2	0.4	0	0	2	0.3		
Experienced physical violence in previous 12 months									
	Yes	64	11.9	47	46.1	111	17.0	14.0	9.5-18.5
	No	471	87.7	55	53.9	526	80.7	85.7	81.2-90.3
	Don't know/missing	2	0.4	0	0	2	0.3		
Was forced into sex in previous 12 months									
	Yes	29	5.4	18	17.6	47	7.2	6.6	3.3-10.0
	No	508	94.6	84	82.4	592	90.8	93.4	90.0-96.7
	Don't know/missing	0	0	0	0	0	0		

Has ever been to jail/prison									
Yes	156	29.1	43	42.2	199	30.5	27.6	21.4-33.9	
No	380	70.8	59	57.8	439	67.3	72.4	66.1-78.6	
Refused	1	0.2	0	0	1	0.2			

Mental Health

Suggested anxiety and depression were high in MSM and TGW (Table 9). Based on the complete PHQ-4 scale, 46.7% of MSM and 72.3% of TGW had scores above for anxiety and depression. Self-harm and suicidal ideation were likewise high among MSM and TGW, with 8.9% of MSM and 21.6% of TGW.

Table 9: Mental Health Among MSM and TGP

		MSM (n=537)		TGW (n=102)		Total (n=652)			
		Crude		Crude		Crude		RDS-Adjusted	
		n	%	n	%	n	%	%	95% CI
Anxiety measure from PHQ-4									
Anxiety suggested	99	18.4	36	31.0	135	20.7	17.2	10.2-24.2	
No anxiety suggested	436	81.2	79	68.1	515	79.0	82.8	75.8-89.8	
Missing	2	0.4	1	0.9	3	0.5			
Depression measure from PHQ-4									
Depression suggested	79	14.7	32	27.6	111	17.0	14.4	6.0-22.8	
No depression suggested	457	85.1	84	72.4	541	83.0	85.6	77.2-94.0	
Missing	1	0.2	0	0	1	0.2			
Combined PHQ-4 anxiety and depression score									
Normal	283	52.7	31	26.7	314	48.2	56.5	49.8-63.4	
Mild	180	33.5	57	49.1	237	36.3	31.6	25.5-37.6	
Moderate	59	11.0	23	19.8	82	12.6	9.6	6.3-12.9	
Severe	12	2.2	4	3.4	16	2.5	2.2	0.7-3.7	
Missing	3	0.6	1	0.9	4	0.6			
Self-harm thoughts in previous 12 months									
Yes	156	29.1	50	49.0	211	32.4	31.0	26.9-35.2	
No	381	70.9	52	51.0	441	67.6	69.0	64.8-73.1	
Self-harm thoughts in previous 4 weeks									
Yes	104	19.4	34	33.3	140	21.5	20.7	17.3-24.0	
No	433	80.6	68	66.7	512	78.5	79.3	76.0-82.7	
Attempted self-harm in Previous 6 Months									
Yes	48	8.9	22	21.6	72	11.0	11.3	8.6-14.0	
No	489	91.1	80	78.4	580	89.0	88.7	86.0-91.4	

Access to Services

A majority of MSM and TGW (75.2% and 73.5%, respectively) received HIV prevention information in the previous year (Table 10). Fewer were aware of any civil society or religious organizations that deliver non-medical services to the community (39.7% of MSM and 56.9% of TGW). Attending meetings or programs related to HIV in the six months preceding the survey was reported by 48.8% of MSM and 62.7% of TGW. About one third of MSM (35.8%) and TGW (34.3%) had contact with a peer educator in the six months preceding the survey.

Table 10: Service Provision among MSM and TGP

		MSM (n=537)		TGW (n=102)		Total (n=652)			
		Crude		Crude		Crude		RDS-Adjusted	
		n	%	n	%	n	%	%	95% CI
Received any information about how to prevent transmission of HIV from someone at the Ministry of Health or any other health service organization in previous 12 months									
	Yes	404	75.2	75	73.5	483	74.1	74.7	72.0-77.4
	No	133	24.8	25	24.5	167	25.6	25.0	22.6-28.0
	Don't know/missing	0	0	2	2.0	2	0.3		
Aware of any civil society or religious organization(s) that deliver non-medical assistance or advice to people who are MSM/Transgender									
	Yes	213	39.7	58	56.9	276	42.3	42.5	39.1-46.0
	No	319	59.4	43	42.2	370	56.7	57.5	54.0-60.9
	Don't know/missing	5	0.9	1	1.0	6	0.9		
Attended any meetings or programs to discuss HIV and/or AIDS in previous 6 months									
	Yes	262	48.8	64	62.7	329	50.5	50.2	47.0-53.5
	No	273	50.8	38	37.3	321	49.2	49.8	46.5-53.0
	Don't know/missing	2	0.4	0	0	2	0.3		
Contact with a health peer educator in the community in previous 6 months									
	Yes	192	35.8	35	34.3	229	35.1	36.4	32.7-40.1
	No	344	64.1	66	64.7	421	64.6	63.6	59.9-67.3
	Don't know/missing	1	0.2	1	1.0	2	0.3		

Healthcare Utilization

Substantial proportions of MSM (31.7%) and TGW (38.2) did not access medical care in the last 12 months (Table 11). The majority who did receive healthcare visited a public clinic or hospital (68.7% of MSM and 74.2% of TGW). Of those who sought care, difficulties in accessing care was reported by 14.2% of MSM and 24.2% of TGW.

Table 11: Healthcare Utilization among MSM and TGP

		MSM (n=537)		TGW (n=102)		Total (n=652)			
		Crude		Crude		Crude		RDS-Adjusted	
		n	%	n	%	n	%	%	95% CI
Received Medical Care in Previous 12 Months									

Yes	367	68.3	62	60.8	439	67.3	67.3	64.6-70.1
No	170	31.7	39	38.2	212	32.5	32.7	29.9-35.4
Missing		0		0	1	0.2		
Health Center Last Visited of Those Received Care								
AHF	2	0.5	1	1.6	3	0.7	0.4	<.01-1.6
JASL	48	13.1	7	11.3	57	13.0	12.8	9.8-16.0
Pharmacy	1	0.3	0	0	1	0.2	0.2	<.01-1.1
Public Clinic/Hospital	252	68.7	46	74.2	304	69.2	74.5	68.8-80.2
Private	33	9.0	1	1.6	35	8.0	8.8	6.6-10.9
Other	8	2.2	3	4.8	11	2.5	3.3	<.01-9.4
Missing/Not listed	23	6.3	4	6.5	28	6.4		
Had Difficulty in Accessing Healthcare of Those who Accessed								
Yes	52	14.2	15	24.2	69	15.7	14.7	10.3-19.0
No	315	85.8	47	75.8	370	84.3	85.3	81.0-89.7

HIV Knowledge

Most MSM and TGW participants were able to correctly answer individual HIV knowledge indicators. By composite score, 61.8% of MSM and 48.0% of TGW correctly answered all five questions.

Table 12: HIV Knowledge among MSM and TGP

	MSM (n=537)		TGW (n=102)		Total (n=652)			
	Crude		Crude		Crude		RDS-Adjusted	
	n	%	n	%	n	%	%	95% CI
HIV knowledge questions correct responses								
1. Can having sex with only one faithful partner reduce the risk of HIV transmission?	451	84.0	76	74.5	537	82.4	84.1	81.0-87.2
2. Can using condoms reduce the risk of HIV transmission?	477	88.8	81	79.4	568	87.1	86.5	83.4-89.6
3 Can a healthy-looking person have HIV?	520	96.8	95	93.1	628	96.3	96.6	94.9-98.3
4. Can a person get HIV from Mosquito bites?	451	84.0	88	86.3	548	84.0	84.5	81.4-87.6
5. Can a person get HIV by sharing a meal with someone who is infected?	471	87.7	91	89.2	573	87.9	89.4	86.8-91.9
Composite score of all 5 correct	332	61.8	49	48.0	388	59.5	61.2	58.2-64.2

HIV Testing

Most MSM (95.0%) and TGW (97.1%) had tested for HIV at least once in their lifetimes (Table 13), with 50.5% of MSM and 82.0% of TGW having most recently tested within the last six months. Testing in the last 12 months was reported by 79.7% of MSM and 77.0% of TGW.

One-fifth of MSM (19.9%) and 6.9% of TGW reported having heard of self-testing, with few having ever used a self-test (2.0% and 1.0%, respectively). At least some willingness to use a self-test was reported by 64.4% of MSM and 60.8% of TGW. Notably, 18.0% of MSM and 27.3% of TGW did not reveal their MSM and TGW status to counselors at the time of last HIV test.

Table 13: HIV Testing Behavior among MSM and TGP									
		MSM (n=537)		TGW (n=102)		Total (n=652)			
		Crude		Crude		Crude		RDS-Adjusted	
		n	%	n	%	n	%	%	95% CI
Previous tested for HIV									
	Yes	510	95.0	99	97.1	619	94.9	94.5	92.4-96.6
	No	26	4.8	3	2.9	31	4.8	5.5	3.4-7.6
	Missing	1	0.2	0	0	2	0.3		
Time since last HIV test among HIV Negative and Unknown Infection									
	0-3 months	195	45.6	20	27.0	217	42.2	43.0	38.0-48.1
	4-6 months	76	17.8	21	28.4	100	19.5	22.8	18.6-26.8
	7 months - 1 year	70	16.4	16	21.6	86	16.7	18.1	14.2-22.1
	Greater than 1 year	59	13.8	14	18.9	76	14.8	16.1	12.4-19.9
	Never tested	26	6.1	3	4.1	31	6.0	--	--
	Missing	2	0.5	0	0.0	4	0.8	--	--
Time since last HIV test among HIV negative and unknown infection									
	<=12 Months	341	79.7	57	77.0	403	78.4	83.9	80.1-87.7
	Greater than 1 year	59	13.8	14	18.9	76	14.8	16.1	12.3-19.9
	Missing/never tested	28	6.6	3	4.1	35	6.8	--	--
Location of Last Test									
	Government facility	143	26.6	32	31.4	178	27.3	30.1	25.8-34.4
	Mobile counseling	133	24.8	22	21.6	159	24.4	24.8	20.3-29.3
	NGO	170	31.7	38	37.3	210	32.2	34.2	30.5-37.9
	Private clinic	35	6.5	1	1.0	37	5.7	5.3	3.7-6.9
	School	7	1.3	1	1.0	8	1.2	1.1	<.01-2.4
	Work	1	0.2	0	0	1	0.2	0.1	<.01-0.6
	Other	20	3.7	5	4.9	25	3.8	4.3	1.1-7.5
	Never tested	26	4.8	3	2.9	31	4.8	--	--
	Missing	1	0.2	0	0	3	0.5	--	--
Heard of HIV self-testing									
	Yes	107	19.9	7	6.9	117	17.9	21.5	15.6-27.4
	No	430	80.1	95	93.1	535	82.1	78.5	72.6-84.4
	Missing	0	0	0	0	0	0		
Used HIV self testing									
	Yes	11	2.0	1	1.0	13	2.0	1.5	<.01-3.0
	No	526	98.0	101	99.0	637	97.7	98.5	97.0->99.9

Willingness to use HIV self-test in the future									
	Very unlikely	76	14.2	17	16.7	95	14.6	15.3	10.3-20.2
	Somewhat unlikely	27	5.0	3	2.9	31	4.8	3.2	1.5-4.9
	Neutral	80	14.9	18	17.6	102	15.6	16.8	11.5-22.0
	Somewhat likely	101	18.8	22	21.6	124	19.0	17.0	11.5-22.5
	Very likely	245	45.6	40	39.2	290	44.5	46.0	39.1-53.0
	Don't know/missing	8	1.5	2	2.0	10	1.5	--	--
Revealed Key Population Status to Last HIV Test of those tested									
	Yes	417	81.8	71	71.7	493	79.6	79.9	76.3-83.4
	No	92	18.0	27	27.3	124	20.0	20.1	16.6-23.7
	Missing	1	0.2	1	1.0	2	0.3		

PrEP Awareness, Willingness, and Use

Table 14 describes PrEP indicators based on the sub-sample of those who tested HIV-negative and those who thought they were HIV-negative (i.e., would be potentially eligible for PrEP). Among this group of MSM, 43.9% were aware of PrEP. Among their TGW counterparts, 25.3% were aware of PrEP. After PrEP was described to participants, 73.3% of potentially eligible MSM and 58.7% of TGW were at least somewhat willing to use PrEP. Preferred sites to receive PrEP tended to be outside of HIV care clinics, particularly at health centers or through NGO or health department outreach.

Table 14: PrEP Knowledge and Use among HIV-Negative and Unaware HIV-Positive MSM and TGW									
		MSM (n=428)		TGW (n=75)		Total (n=514)			
		Crude		Crude		Crude		RDS-Adjusted	
		n	%	n	%	n	%	%	95% CI
Aware of PrEP									
	Yes	188	43.9	19	25.3	210	40.9	48.5	43.5-53.4
	No	200	46.7	32	42.7	237	46.1	51.5	46.6-56.5
	Missing	40	9.3	24	32.0	67	13.0		
Willingness to Use PrEP									
	Very willing	230	53.7	34	45.3	267	51.9	61.2	56.5-66.0
	Moderately willing	43	10.0	5	6.7	48	9.3	9.7	5.9-13.5
	Somewhat willing	41	9.6	5	6.7	46	8.9	11.6	6.9-16.3
	Not willing	68	15.9	7	9.3	79	15.4	17.5	13.2-21.8
	Missing	46	10.7	24	32.0	74	14.4		
Where Participant Prefers to Get PrEP (multiple response)									
	HIV care clinic	41	15.0	5	12.8	46	14.6	14.8	9.5-20.1
	Health center, NOT HIV care clinic	196	71.8	19	48.7	216	68.6	70.7	65.1-76.2

NGO or health department outreach	111	40.7	24	61.5	137	43.5	43.6	37.9-49.1
Peers	31	11.4	2	5.1	33	10.5	9.5	4.8-14.3
Other source	26	9.5	9	23.1	36	11.4	11.5	6.2-16.7

Syphilis and HIV Prevalence

Syphilis prevalence was 30.5% antibody positive among MSM and 43.1% among TGW (Table 15). HIV prevalence was 29.6% among MSM and 51.0% among TGW.

Table 15: Syphilis Test Results and HIV Prevalence									
		MSM (n=537)		TGW (n=102)		Total (n=652)			
		Crude		Crude		Crude		RDS-Adjusted	
		n	%	n	%	n	%	%	95% CI
Syphilis Test Result									
	Positive	164	30.5	44	43.1	210	32.2	31.2	27.0-35.3
	Negative	373	69.5	58	56.9	442	67.8	68.8	64.7-73.0
	Missing	0	0	0	0	0	0		
HIV rapid test results									
	HIV Positive	159	29.6	52	51.0	213	32.7	32.9	28.2-37.7
	HIV Negative	374	69.6	50	49.0	435	66.7	67.1	62.3-71.8
	Missing	4	0.7	0	0	4	0.6	--	--

Correlates of HIV Infection

Table 16 shows correlates of HIV infection. Please refer to appendix 2 for bivariate results per indicator.

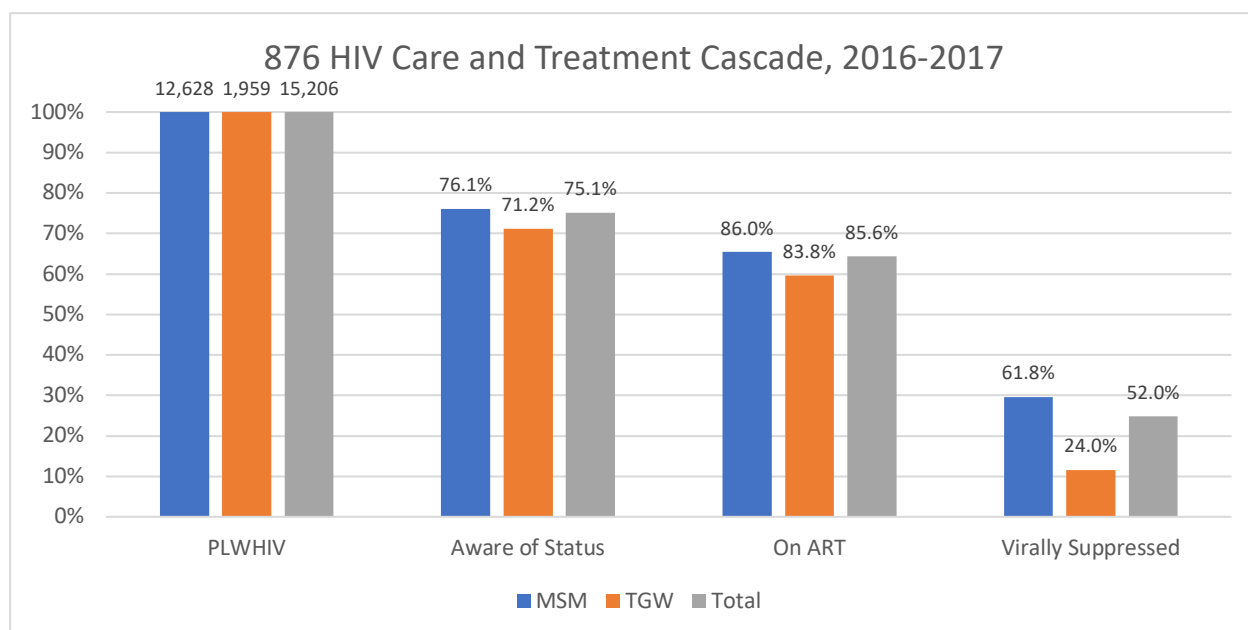
Table 16: Correlates of HIV Infection				
		sig	Adjusted OR	95% CI
Sexual Identity				
	MSM/TGM	REF		
	TGW	<.001	2.7	1.6-4.5
Age				
	16-24	REF		
	25 and older	<.001	2.9	1.9-2.7
Any reported STI symptoms in previous 12 months				
	No	REF		
	Yes	<.001	1.7	1.1-2.7
Syphilis rapid test result				
	Syphilis non-reactive	REF		
	Syphilis reactive	<.001	4.5	3.0-6.8
Last tested at government facility				
	No	REF		

Yes	<.001	3.1	2.0-4.8
Tested for HIV within one year			
No	REF		
Yes	<.001	0.3	0.2-0.5

TGW had 2.7 (1.6-4.5) times the odds of HIV infection of MSM. Participants age 25 and older had 2.9 (1.9-2.7) times the odds of HIV infection than younger participants. Participants who reported having any STI symptoms (ulcers, warts, discharge), had 1.7 (1.1-2.7) times the odds of HIV compared to those not reporting symptoms. Having had a reactive syphilis rapid test was the strongest predictor of HIV infection, with those testing positive have 4.5 (3.0-6.8) times the odds to test HIV-positive. No co-linearity was observed between any reported STI symptoms and syphilis rapid test result. We also observed an association between those who last tested at a government clinic and being HIV-positive [OR 3.1 (2.0-4.8)]. The only significant protective association was found among those who had tested for HIV in the previous year [OR 0.3 (0.2-0.5)] compared to those who did not test or tested greater than one year.

Continuum of Engagement in HIV Care (Cascade)

Figure 5: HIV Care and Treatment Cascade



The number of participants per key population estimated to be living with HIV are given above the first bars. Of the 42,375 MSM, we estimate that there are 12,628 living with HIV. For TGW we estimate 1,959 are living with HIV. For MSM, 76.1% of those living with HIV are aware of

their status, while 71.2% of TGW are aware. Out of those who are aware, 86.0% of MSM and 83.8% of TGW are estimated to be on ART. Viral suppression for those on ART are 61.8% for MSM and 24.0% for TGW.

Global AIDS Monitoring (GAM) Indicators

Table 17: GAM Indicators by Population, 876 Study 2018			
GAM #	Indicator	MSM	TGW
3.3	HIV Prevalence	29.60%	51.00%
3.4	HIV Testing	85.70%	80.80%
3.5	ART coverage	50.90%	26.90%
3.6	Condom Use	71.40%	60.80%
3.7	Prevention coverage	22.70%	34.30%

Table 18: GAM Indicators by Population and Age, 876 Study 2018			
	MSM	TGW	Total
16 – 19 yrs	10.70%	35.00%	14.60%
20 – 24 yrs	23.40%	41.70%	26.30%
25+ yrs	42.60%	65.20%	45.60%
All ages	29.60%	51.00%	32.80%
% Diagnosed (1 st 90 -all ages)	68.50%	51.90%	64.50%

Population Size of MSM and TGW

Several methods were used to arrive at estimates of the number of MSM and TGW in Jamaica. The convergence or central tendency (median) of the results of the several methods was used as the final estimates.

Men Who Have Sex with Men

Population size estimates by method:

- Literature review. Total general population size of Jamaica = 961,085 adult males; literature extrapolation (prior) = 46,613 (28,832 – 64,392)
- Unique object multiplier = not available due to errors in distribution to population
- Service multiplier = 4,192 (3,637 – 4,888)
- Wisdom of the crowd = 30,000 (20,000 – 40,000)

Anchored multiplier (variance adjusted) combined estimate: 42,375 (95% CI: 28,278 – 58,855)

Proportion likely to disclose as MSM: 34,705 (95% CI: 33,163 - 36,080)

Estimates for the reachable population:

Approach 1: 'The Already Reached' ... Unduplicated count of MSM already reached by existing outreach services (Jan – Dec 2017)			
			5,783
Approach 2: 'The Need to be Reached' ... Estimated count of those who need to be reached by outreach services.			
Indicator	Programmatic unmet need	Sample proportion	Count
<i>Any episode of unprotected (insertive) anal intercourse</i>	Condom promotion		
Disclosed as MSM		24.3%	10,297
Not disclosed as MSM		5.6%	2,373
<i>Any episode of unprotected (receptive) anal intercourse</i>	Condom promotion		
Disclosed as MSM		25.6%	10,848
Not disclosed as MSM		3.7%	1,568
<i>Low knowledge of HIV transmission</i>	Information Education Communication		
Disclosed as MSM		29.6% ¹	12,543
Not disclosed as MSM		7.7%	3,262
<i>Not tested for HIV within past 6 months</i> ²	Testing		
Disclosed as MSM		28.0%	11,865
Not disclosed as MSM		8.7%	3,687
<i>Unknown HIV positive</i> ³	Diagnosis		
Disclosed as MSM		20.8%	8,814
Not disclosed as MSM		9.4%	3,983

¹ This denominator is the total number of people in the sample who answered this question. The denominator may not always sum to the complete sample size.

² Doesn't include participants who never saw a provider.

³ Doesn't include participants who never saw a provider.

Transgender Women

Population size estimates:

- General population size = 961,085 adult males; literature extrapolation (prior) = 4,805 (3,844 – 5,767)
- Wisdom of the crowd = 800 (500 – 1,200)
- Service multiplier = 222 (170 – 304)
- Unique object multiplier = not available due to errors in distribution to population

Anchored multiplier (variance adjusted) combined estimate: 3,841 (95% CI: 3,142 – 4,646)

Proportion likely to disclose as TGW: 2,781 (2,401 - 3,111)

Estimates for the reachable population:

Approach 1: 'The Already Reached' ... Unduplicated count of TGW already reached by existing outreach services (Jan – Dec 2017)			
			317
Approach 2: 'The Need to be Reached' ... Estimated count of those who need to be reached by outreach services.			
Indicator	Programmatic unmet need	Sample proportion	Count
<i>Any episode of unprotected (insertive) anal intercourse</i>	Condom promotion		
Disclosed as TGW		10.8%	415
Not disclosed as TGW		1.0%	38
<i>Any episode of unprotected (receptive) anal intercourse</i>	Condom promotion		
Disclosed as TGW		35.3%	1,356
Not disclosed as TGW		13.7%	526
<i>Low knowledge of HIV transmission</i>	Information Education Communication		
Disclosed as TGW		36.7% ¹	1,410
Not disclosed as TGW		14.3%	549
<i>Not tested for HIV within past 6 months²</i>	Testing		
Disclosed as TGW		31.6%	1,214
Not disclosed as TGW		12.2%	469
<i>Unknown HIV positive³</i>	Diagnosis		
Disclosed as TGW		28.6%	1,099
Not disclosed as TGW		16.3%	626

¹ This denominator is the total number of people in the sample who answered this question. The denominator may not always sum to the complete sample size.

² Doesn't include participants who never saw a provider.

³ Doesn't include participants who never saw a provider.

Sub-National Estimates:

Sub-national estimates were calculated based on the distribution of the 876 Study sample.

Region	MSM	TGW
Western Regional Health Authority	12,967	453
Northeastern Regional Health Authority	3,729	676
Southeastern Regional Health Authority	22,374	2,032
Southern Regional Health Authority	3,305	676

DISCUSSION

The 876 Study successfully met all proposed objectives, including the primary objective of estimating HIV prevalence in Jamaica (at 29.6% among MSM and 51.0% among TGW). The study was the first to use RDS in Jamaica, recruiting a sample that was diverse with respect to age, gender identity, sexual orientation, parish of residence, education, and employment. This study recruited one of the largest samples of MSM and the largest sample of TGW in the Caribbean to date. This study was also the first to include TGM, highlighting a previously undocumented diversity in gender identity of persons in Jamaica. We reached participants from every parish in Jamaica including many who chose to complete the study at an office far from their residence. This suggests that these populations are highly mobile, interconnected, and willing to engage in public health surveillance. A summary of key findings by objective and recommendations follows.

Objective 1: Determine the prevalence of HIV and assess the potential associations between HIV prevalence and risk behaviors among MSM and TGP in Jamaica

MSM and TGW in Jamaica have a very high burden of HIV as confirmed by this large, population-based RDS survey. HIV prevalence differed between the populations, with 29.6% of MSM and 51.0% of TGW testing positive for HIV. The very high prevalence among TGW is consistent with literature from around the world, placing this group among the most at-risk in the region, and calling for specifically tailored programs for this population.

Data gleaned from indicators of sexual behavior help understand drivers of HIV infection and guide effective prevention programs. Both populations reported young ages of sexual debut, suggesting that very young MSM and TGW must be included in prevention programming. MSM and TGW reported similar venues for finding sexual partners, which provide key locations for reaching these populations with prevention messaging via the internet, through peers, and at social gatherings. Condom use at last sex was low for MSM and TGW. Roughly one-third of participants reported condoms being 'somewhat easy to access' suggesting that condoms distribution could be improved. PrEP awareness was also low in both groups with only one quarter of TGW being aware. Once informed about PrEP, only half of participants stated they would be willing to use PrEP. More work needs to be done to sensitize the populations to the benefits of PrEP, create demand among those at risk, and identify channels for distribution.

Key correlates of HIV infection were found to be having a reactive syphilis test, STI symptoms in the previous 12 months, TGW status, age of over 25, and testing at a government facility, with testing for HIV in the previous 12 months associated with lower prevalence. The comorbidity of STI and HIV and the behavioral link between acquiring these infections has been well established in the literature. The longer period of being at risk for HIV among older participants likely explains the association with age over 25 years. Nonetheless, infection was present in the youngest age group, indicating risk starting at a very young age and the need for prevention programs to reach the very young to prevent infection. As for those who tested at a government facility being at higher risk, we hypothesize a potential bias of confirmatory testing being done at government facilities upon linkage to care, though it could also be that marginalized groups within these populations are more likely to test at a government clinic. Lastly, we observed a protective effect among those who have tested for HIV within the past year, supporting regular testing as an essential prevention strategy.

Objective 2: Assess engagement along the continuum of HIV prevention and care among MSM and TGP

Though MSM and TGP reported high levels of ever testing for HIV, regular testing was low. Over 40% are estimated to not be in compliance with recommended HIV testing every six months, and over one fifth had not tested within the last year or never tested at all. A majority of participants test at government sites, though with members of the population not disclosing their MSM/TGP status, they may not be receiving appropriate risk assessment and recommendations for re-testing. A majority of participants, once sensitized to HIV self-test kits, were very or somewhat willing to use them in the future. HIV self-test kits provide a safe, truly anonymous environment in which to conduct a test and may help to increase the proportion of MSM and TGP who test regularly. The low proportion of people testing regularly is reflected in the proportion of people living with HIV estimated to be aware of their status (76.1% of MSM; 71.2% of TGP) – well below the first of the 90-90-90 targets. For those aware of their status, most are on ART (86.0% of MSM; 83.8% of TGP). Though this is also below 90-90-90 targets. Viral suppression, specifically among TGP, is extremely low, suggesting either non-adherence to medication or these key populations falling out of care.

Objective 3: Estimate the population size of MSM and TGP

The estimates for the number of MSM and TGP from our study are consistent with what has been published in the past for Jamaica. The MSM population is estimated to be 42,375 (95% CI: 28,278 – 58,855), and TGP 3,841 (95% CI: 3,142 – 4,646). These updated population size estimates will be useful in setting targets for programs focused on key populations.

Objective 4: Assess the psychosocial needs of MSM and TGP

Indicators of unmet psychosocial and mental health needs were high in our study and particularly so for TGP. Almost three-fourths of TGP had suggested mild to severe anxiety and depression as measured by the PHQ-4. Concurrently, almost half of TGP reported having self-harm thoughts in the preceding 12 months, with one third having had such thoughts in the previous four weeks. Over one-fifth of TGP had attempted self-harm in the previous six months, suggesting high levels of unaddressed mental stress in this population. MSM and TGP reported high levels of stigmatizing behavior, including verbal assault. TGP experienced more extreme assault, with almost half having experienced physical violence and nearly one in five having been raped in the previous year.

Almost a third of MSM were designated as being hazardous drinkers from the AUDIT-C. TGP reported less drinking than MSM, but over one-fourth were still classified as hazardous. Little use of non-prescribed drugs was observed, with no participant acknowledging injection drug use. Marijuana was used by almost half of participants with more TGP reporting use in the preceding month.

Objective 5: Recruit a cohort for future HIV incidence surveillance and engagement in care among MSM and TGP in Jamaica.

Participants were separately invited to give their name and contact information for MOH to use in a future cohort study. In the event, most participants consented showing a willingness and acceptability for engaging in research by Jamaican MSM and TGW.

Objective 6: Build local capacity to conduct BSS and population size estimation among key populations in Jamaica

Study staff at each of the three sites were Jamaican and were trained in BSS methods and procedures. The team showed an extremely high level of professionalism and care when working with the marginalized population. Most study staff work in the health services sector in Jamaica, and now have a great technical expertise for conducting a BSS study. In their roles as peer navigators, HIV counselors and testers, and coordinators, they gained essential skills in working sensitively and compassionately with these populations. The investigators commend the outstanding job completed by staff and highlight their essential role in contributing to the success of the study.

Objective 7: Strengthen syphilis reporting, referral, and linkage to treatment services.

As with HIV study prevalence, we also noted a high number of reactive tests for syphilis. Though the rapid test was unable to detect current infection, participants with reactive results were referred to sites for follow-up care. As we noted the high correlation between syphilis and HIV infection, it is essential that routine STI screening and immediate treatment initiation be a part of any prevention/treatment program with these key populations.

Implications and recommendations:

1. **Combination Prevention for Very Young MSM.** The HIV prevalence of 29.6% among MSM in the current sample is consistent with the prevalence of 31.4% reported in the 2011 IBBSS conducted in Jamaica (MOH, 2015), which notably included an unknown number of TGW. The substantial prevalence of HIV among younger MSM points to ongoing transmission. A majority of the MSM in this study (62.9%) reported sexual debut by age 15 years, and almost one in four are infected by age 25. The comprehensive package of HIV prevention services that is currently offered to MSM should therefore be reviewed to identify a tailored combination for young MSM.

- a) This comprehensive package should be based on youth-friendly behavioral, biomedical, and structural interventions that address the needs of young MSM. Efforts to review policies that limit access to sexual and reproductive health for young people should remain a priority to ensure key health interventions are accessible for all sexually active youth. Combination prevention activities should include behavioral and biomedical prevention interventions with evidence of efficacy (including PrEP), mental health and substance use services, access to frequent HIV and syphilis testing, and peer or youth-oriented navigation for those who test HIV positive to increase linkage, adherence, and retention to achieve viral suppression.
- b) In addition to traditional approaches of reaching young MSM, activities in the comprehensive package should be delivered in schools, on internet-based platforms (i.e., social media), and through peer / social networks.
- c) Increasing PrEP awareness, demand among those eligible, and modes of delivery are a priority. Most (73.3%) of potentially eligible MSM were at least somewhat willing to use PrEP. To increase acceptance and adherence, PrEP services should be provided in settings not usually associated with HIV treatment, such as health-centers that are not HIV treatment sites, and should include screening and treatment for mental disorders. Barriers to youth accessing SRH services at health centers should be removed, and youth-friendly access points for PrEP established.

2. **HIV Prevention Programs for TGW.** The astonishingly high HIV prevalence of 51.0% among TGW in this study highlights a striking disparity in risk and vulnerability for transgender women. The extreme burden that transgender women experience requires programs specifically targeting their prevention and treatment needs.

- a) Experiences of stigma, discrimination, and transphobic violence were high among TGW. TGW reported experiencing refusal of essential services, verbal assault, physical violence, and rape. Interventions addressing their educational, housing, safety, legal, health care, and employment needs may mitigate the impact of stigma and discrimination.
- b) TGW reported experiencing significantly high levels discrimination in health care settings. Designating specific clinics for transgender health as well as sensitizing all public sector staff could increase access to non-discriminatory services, and create the space for persons to receive transgender-specific medical care in

gender-affirming settings. Providers in all clinics should receive training in transgender-specific and transgender-sensitive service delivery.

- c) The high HIV burden in the transgender population coupled with the high rate of risk behaviors (early age of sexual debut, multiple sexual partners, involvement in sex work, receptive anal sex, etc.) reported in this study suggest transgender women are a priority population to receive PrEP in Jamaica. Linkage and access to other programs for sexual-risk reduction, STI screening and treatment, and mental health screening and services may be added to the delivery of PrEP. PrEP programs should also focus on young transgender women (age 15 - 24). More than two in five transgender women between the ages of 20 and 24 were already HIV-positive.
- d) A large proportion of HIV-positive TGW reported that they were newly diagnosed in the study. It is recommended that prevention activities should include opportunities for frequent testing for HIV and syphilis and strategies for personalization of risk, as a starting point for promoting prevention techniques.
- e) Measures of anxiety, depression, and suicidal ideation and attempt revealed significant mental health morbidity among TGW. Interventions that include mental health screening and treatment should be offered at each engagement with transgender women. The use of internet platforms, telephone hotlines, or peer support should be explored as options for delivery. With respect to structural factors, transgender identities are not legally recognized in Jamaica and there are no local guidelines around the provision of medically administered hormones or gender confirmation surgery. Such interventions can significantly reduce mental distress and improve quality of life among TGP.

3. Reaching 90-90-90 targets. The estimated HIV prevention and care continuum for MSM and transgender women reveal several points for improvement. Actions to improve the continuum may include interventions to address:

- a) Targeted testing - innovations for testing can increase the identification of undiagnosed PLHIV. It is recommended that the comprehensive package include strategies to increase self-testing, peer-driven recruitment to testing (such as utilizing RDS-like approaches to gain access to hidden networks), and index testing and partner notification. Policies that create a barrier to access for young MSM and transgender women should be reviewed.
- b) Linkage to care - M&E tools that improve the monitoring of timely linkage to care after testing can provide valuable information on gaps in linkage. Designated MSM-friendly, transgender-friendly, and youth-friendly clinics can increase access for marginalized persons.
- c) Initiating ART - Early initiation of ART is recommended for better clinical outcomes. Same day ART starter packs may be introduced to facilitate timely initiation.
- d) Adherence and viral suppression - timely monitoring of patients on ART can improve outcomes. Adding features to the HIV surveillance system that support appointment reminders, adherence reminders can identify potential defaulters. The national response may also consider the impact that transitioning to

dolutegravir (DTG)-based regimens as the preferred first-line antiretroviral therapy (ART) will have on improving initiation, retention, and viral suppression.

4. Capitalizing on Social Media and Peer Connections. The national HIV response should reevaluate how messages and interventions are delivered. The findings from this study suggest internet and social media might provide greater access to persons and may be utilized to deliver interventions to enhance HIV prevention and treatment. As seen in many locales around the world, participants in this study also reported that the internet is the primary venue for meeting potential sex partners. Internet-based interventions can promote individual and couples testing for HIV and syphilis testing and support linkage and retention in treatment (scheduling appointments, appointment and prescription reminder, adherence monitoring, etc.). Meeting sexual partners through friends was also reported as common, which is expected given the dense social networks across the island. Friends and social networks could potentially provide peer-delivered interventions. Only about one-third of MSM and TGW reported that they had contact with a peer educator in the six months preceding their participation in the survey. This was the first study to use Respondent-Driven Sampling in Jamaica, and it recruited a sample that was diverse in respect of age, gender identity, sexual orientation, parish of residence, education and employment. The study attracted several seeds and participants with extensive networks. The HIV program may capitalize on RDS-like approaches to recruit persons for the prevention and treatment program.

5. On-going Need for Strategic Information among Key Populations. BSS should be repeated every three to five years to monitor trends in behavior and HIV prevalence among MSM and TGP, and progress towards “getting to zero” in Jamaica. Future rounds of BSS can be improved by including:

- a) Additional questions to gain a better understanding of the gender identity and partnership patterns of MSM and TGW. For example, binary gender questions should be replaced with questions that distinguish between cis-gender and trans-gender partners, and include additional non-binary gender responses.
- b) Formative assessment should explore the feasibility and benefits of implementing a BSS specifically for transgender and other non-binary persons, independent of the MSM study.
- c) Additional questions to gain in-depth understanding of the unique drivers of risk among transgender participants.
- d) Future BSS should strive for larger samples of TGM to increase the understanding of their sexual and reproductive and HIV prevention needs.
- e) Research designs that facilitate direct measurement of the full cascade, such as sample collection to assess recency of HIV infection, ART exposure, and VL testing.
- f) Direct collection of samples to assess active syphilis through serological testing.
- g) Additional questions to clarify the role of the internet in delivering services, particularly the level of access to websites of local stakeholder and partner.

References

- Figueroa, J. P., S. S. Weir, C. Jones-Cooper, L. Byfield, M. M. Hobbs, I. McKnight, and S. Cummings. 2013. "High HIV Prevalence among Men Who Have Sex with Men in Jamaica Is Associated with Social Vulnerability and Other Sexually Transmitted Infections." *The West Indian Medical Journal* 62 (4): 286–91. <https://doi.org/10.7727/wimj.2011.207>.
- Figueroa, J. P., Carol Jones Cooper, Jessie K. Edwards, Lovette Byfield, Shashauna Eastman, Marcia M. Hobbs, and Sharon S. Weir. 2015. "Understanding the High Prevalence of HIV and Other Sexually Transmitted Infections among Socio-Economically Vulnerable Men Who Have Sex with Men in Jamaica." Edited by Joan A Caylà. *PLOS ONE* 10 (2): e0117686. <https://doi.org/10.1371/journal.pone.0117686>.
- Johnston, Lisa. 2013. "Technical Report. 2013. Introduction to Respondent-Driven Sampling." World Health Organization.
- Maltiel, Rachael, Adrian E. Raftery, Tyler H. McCormick, and Aaron J. Baraff. 2015. "Estimating Population Size Using the Network Scale up Method." *The Annals of Applied Statistics* 9 (3): 1247–77. <https://doi.org/10.1214/15-AOAS827>.
- Ministry of Health. 2015. "The Jamaica National Integrated Strategic Plan for Sexual and Reproductive Health and HIV (NISP) 2014-2019." <https://www.moh.gov.jm/wp-content/uploads/2016/05/Final-National-Integrated-Strategic-Plan-2014-2019-with-SRH-Validation.pdf>.
- Ministry of Health. 2017. "Global AIDS Monitoring Report. 2017 NFPB- SRHA." http://www.unaids.org/sites/default/files/country/documents/JAM_2017_countryreport.pdf.
- Okal, Jerry, Scott Geibel, Nicolas Muraguri, Helgar Musyoki, Waimar Tun, Dita Broz, David Kuria, Andrea Kim, Tom Oluoch, and H Fisher Raymond. 2013. "Estimates of the Size of Key Populations at Risk for HIV Infection: Men Who Have Sex with Men, Female Sex Workers and Injecting Drug Users in Nairobi, Kenya." *Sexually Transmitted Infections* 89 (5): 366–71. <https://doi.org/10.1136/sextrans-2013-051071>.
- Raymond, H Fisher, Sylvia Bereknyei, Nancy Berglas, Jennifer Hunter, Norah Ojeda, and Willi McFarland. 2013. "Estimating Population Size, HIV Prevalence and HIV Incidence among Men Who Have Sex with Men: A Case Example of Synthesising Multiple Empirical Data Sources and Methods in San Francisco." *Sexually Transmitted Infections* 89 (5): 383–87. <https://doi.org/10.1136/sextrans-2012-050675>.
- STATIN. 2017. "End of Year Population by Age and Sex." http://statinja.gov.jm/demo_socialstats/newEndofYearPopulationbyAgeandSex2008.aspx.
- Wesson, Paul D, Ali Mirzazadeh, and Willi McFarland. 2018. "A Bayesian Approach to Synthesize Estimates of the Size of Hidden Populations: The Anchored Multiplier." *International Journal of Epidemiology* 47 (5): 1636–44. <https://doi.org/10.1093/ije/dyy132>.

Appendix A: Selected Tables by Adolescent Age Groups

Table 2a: Demographics of MSM and TGW Among Adolescents													
		MSM				TGW				Total			
		16-19 (n=103)		20-24 (n=192)		16-19 (n=20)		20-24 (n=36)		16-19 (n=123)		20-24 (n=228)	
		n	%	n	%	n	%	n	%	n	%	n	%
Sex at Birth													
	Male	103	100.0%	192	100.0%	20	100.0%	36	100.0%	123	100.0%	228	100.0%
	Female	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Current Gender													
	Male	88	85.4%	171	89.1%	0	0.0%	0	0.0%	88	71.5%	171	75.0%
	Female-to-male	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
	Male-to-female	0	0.0%	0	0.0%	20	100.0%	36	100.0%	20	16.3%	36	15.8%
	Genderqueer	15	14.6%	18	9.4%	0	0.0%	0	0.0%	15	12.2%	18	7.9%
	Other	0	0.0%	3	1.6%	0	0.0%	0	0.0%	0	0.0%	3	1.3%
Sexual Orientation													
	Gay/Lesbian	39	37.9%	97	50.5%	18	90.0%	30	83.3%	57	46.3%	127	55.7%
	Bisexual	64	62.1%	94	49.0%	2	10.0%	3	8.3%	66	53.7%	97	42.5%
	Heterosexual/Straight	0	0.0%	1	0.5%	0	0.0%	3	8.3%	0	0.0%	4	1.8%
	Missing	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Highest education level completed													
	Primary or less	1	1.0%	1	0.5%	0	0.0%	0	0.0%	1	0.8%	1	0.4%
	Secondary	89	86.4%	115	59.9%	12	60.0%	21	58.3%	101	82.1%	136	59.6%
	University or higher	3	2.9%	16	8.3%	0	0.0%	1	2.8%	3	2.4%	17	7.5%
	Vocational	10	9.7%	59	30.7%	8	40.0%	14	38.9%	18	14.6%	73	32.0%
	Other			1	0.5%	0	0.0%	0	0.0%	0	0.0%	1	0.4%
Current student status													
	Full-time student	50	48.5%	29	15.1%	5	25.0%	4	11.1%	55	44.7%	33	14.5%
	Part-time student	3	2.9%	9	4.7%	0	0.0%	4	11.1%	3	2.4%	13	5.7%
	Not a student	50	48.5%	154	80.2%	15	75.0%	28	77.8%	65	52.8%	182	79.8%

Table 3a: Income, Occupation, and Parish of Residence of Adolescent MSM and TGW													
		MSM				TGW				Total			
		16-19 (n=103)		20-24 (n=192)		16-19 (n=20)		20-24 (n=36)		16-19 (n=123)		20-24 (n=228)	
		n	%	n	%	n	%	n	%	n	%	n	%
Household Monthly Income													
	<=40,000 JMD	31	30.1%	80	41.7%	10	50.0%	17	47.2%	41	33.3%	97	42.5%
	41k-149K JMD	30	29.1%	61	31.8%	2	10.0%	9	25.0%	32	26.0%	70	30.7%
	150k-299k JMD	5	4.9%	17	8.9%	0	0.0%	2	5.6%	5	4.1%	19	8.3%
	300k-499k JMD	3	2.9%	5	2.6%	0	0.0%	2	5.6%	3	2.4%	7	3.1%
	>=450k JMD	4	3.9%	1	0.5%	0	0.0%	0	0.0%	4	3.3%	1	0.4%
	Do not know	25	24.3%	23	12.0%	7	35.0%	5	13.9%	32	26.0%	28	12.3%
	Declined	4	3.9%	5	2.6%	1	5.0%	1	2.8%	5	4.1%	6	2.6%
Food Insecurity: Skipped ore reduced meals due to lack of money													
	Yes	22	21.4%	59	30.7%	5	25.0%	13	36.1%	27	22.0%	72	31.6%
	No	81	78.6%	133	69.3%	15	75.0%	23	63.9%	96	78.0%	156	68.4%
Occupation													
	Not employed	74	71.8%	66	34.4%	13	65.0%	19	52.8%	87	70.7%	85	37.3%
	Professional	8	7.8%	38	19.8%	1	5.0%	5	13.9%	9	7.3%	43	18.9%
	Laborer	2	1.9%	11	5.7%	0	0.0%	1	2.8%	2	1.6%	12	5.3%
	Agriculture/fishing	1	1.0%	1	0.5%	1	5.0%	0	0.0%	2	1.6%	1	0.4%
	Driver	0	0.0%	2	1.0%	1	5.0%	0	0.0%	1	0.8%	2	0.9%
	Retail/restaurant/performer	12	11.7%	31	16.1%	2	10.0%	4	11.1%	14	11.4%	35	15.4%
	Security/military/police	2	1.9%	5	2.6%	0	0.0%	0	0.0%	2	1.6%	5	2.2%
	Sex work	0	0.0%	5	2.6%	1	5.0%	3	8.3%	1	0.8%	8	3.5%
	Call center	3	2.9%	27	14.1%	1	5.0%	1	2.8%	4	3.3%	28	12.3%
	Other	0	0.0%	6	3.1%	0	0.0%	3	8.3%	0	0.0%	9	3.9%
	Missing	1	1.0%	0	0.0%	0	0.0%	0	0.0%	1	0.8%	0	0.0%
Parish of Residence													
	Clarendon	8	7.8%	21	10.9%	2	10.0%	8	22.2%	10	8.1%	29	12.7%
	Hanover	5	4.9%	2	1.0%	0	0.0%	0	0.0%	5	4.1%	2	0.9%
	Kingston	25	24.3%	68	35.4%	8	40.0%	11	30.6%	33	26.8%	79	34.6%
	Manchester	1	1.0%	2	1.0%	1	5.0%	0	0.0%	2	1.6%	2	0.9%
	Portland	0	0.0%	4	2.1%	0	0.0%	0	0.0%	0	0.0%	4	1.8%
	St. Andrew	8	7.8%	12	6.3%	2	10.0%	1	2.8%	10	8.1%	13	5.7%
	St. Ann	8	7.8%	10	5.2%	0	0.0%	7	19.4%	8	6.5%	17	7.5%
	St. Catherine	16	15.5%	25	13.0%	4	20.0%	4	11.1%	20	16.3%	29	12.7%
	St. Elizabeth	1	1.0%	1	0.5%	0	0.0%	0	0.0%	1	0.8%	1	0.4%
	St. James	20	19.4%	25	13.0%	0	0.0%	2	5.6%	20	16.3%	27	11.8%
	St. Mary	1	1.0%	1	0.5%	2	10.0%	0	0.0%	3	2.4%	1	0.4%
	St. Thomas	5	4.9%	4	2.1%	0	0.0%	0	0.0%	5	4.1%	4	1.8%
	Trelawney	0	0.0%	3	1.6%	0	0.0%	2	5.6%	0	0.0%	5	2.2%
	Westmoreland	5	4.9%	14	7.3%	1	5.0%	1	2.8%	6	4.9%	15	6.6%

Table 4a: Sexual Behavior of Adolescent MSM and TGW													
		MSM				TGW				Total			
		16-19 (n=103)		20-24 (n=192)		16-19 (n=20)		20-24 (n=36)		16-19 (n=123)		20-24 (n=228)	
		n	%	n	%	n	%	n	%	n	%	n	%
Total number of male partners in previous 12 months													
	0	0	0.0%	2	1.0%	0	0.0%	0	0.0%	0	0.0%	2	0.9%
	1	23	22.3%	41	21.4%	5	25.0%	8	22.2%	28	22.8%	49	21.5%
	2	26	25.2%	43	22.4%	6	30.0%	6	16.7%	32	26.0%	49	21.5%
	3	20	19.4%	27	14.1%	2	10.0%	4	11.1%	22	17.9%	31	13.6%
	4	5	4.9%	18	9.4%	3	15.0%	6	16.7%	8	6.5%	24	10.5%
	5 or more	29	28.2%	61	31.8%	4	20.0%	12	33.3%	33	26.8%	73	32.0%
Total number of female partners in previous 12 months													
	0	54	52.4%	119	62.0%	17	85.0%	32	88.9%	71	57.7%	151	66.2%
	1	19	18.4%	23	12.0%	3	15.0%	1	2.8%	22	17.9%	24	10.5%
	2	10	9.7%	12	6.3%	0	0.0%	1	2.8%	10	8.1%	13	5.7%
	3	6	5.8%	11	5.7%	0	0.0%	1	2.8%	6	4.9%	12	5.3%
	4	4	3.9%	4	2.1%	0	0.0%	0	0.0%	4	3.3%	4	1.8%
	5 or more	9	8.7%	22	11.5%	0	0.0%	1	2.8%	9	7.3%	23	10.1%
	Missing	1	1.0%	1	0.5%					1	0.8%	1	0.4%
Paid for sex with any partners in previous 12 months													
	Yes	7	6.8%	20	10.4%	1	5.0%	3	8.3%	8	6.5%	23	10.1%
	No	96	93.2%	172	89.6%	19	95.0%	33	91.7%	115	93.5%	205	89.9%
Received payment for sex with any partners in previous 12 months													
	Yes	23	22.3%	50	26.0%	6	30.0%	19	52.8%	29	23.6%	69	30.3%
	No	80	77.7%	142	74.0%	14	70.0%	17	47.2%	94	76.4%	159	69.7%
Sexual position preference													
	Receptive	32	31.1%	49	25.5%	14	70.0%	13	36.1%	46	37.4%	62	27.2%
	Insertive	18	17.5%	47	24.5%	1	5.0%	3	8.3%	19	15.4%	50	21.9%
	Oral Only	0	0.0%	2	1.0%	0	0.0%	0	0.0%	0	0.0%	2	0.9%
	Versatile	53	51.5%	92	47.9%	5	25.0%	20	55.6%	58	47.2%	112	49.1%
	None	0	0.0%	1	0.5%	0	0.0%	0	0.0%	0	0.0%	1	0.4%
	Missing	0	0.0%	1	0.5%	0	0.0%	0	0.0%	0	0.0%	1	0.4%
Venues for meeting sexual partners													
	hotel	7	6.8%	13	6.8%	2	10.0%	4	11.1%	9	7.3%	17	7.5%
	street	42	40.8%	80	41.7%	12	60.0%	21	58.3%	54	43.9%	101	44.3%
	friends	52	50.5%	100	52.1%	10	50.0%	22	61.1%	62	50.4%	122	53.5%
	internet	85	82.5%	159	82.8%	17	85.0%	28	77.8%	102	82.9%	187	82.0%
	party	27	26.2%	59	30.7%	3	15.0%	22	61.1%	30	24.4%	81	35.5%
	work	24	23.3%	49	25.5%	4	20.0%	8	22.2%	28	22.8%	57	25.0%
	workshop	18	17.5%	27	14.1%	4	20.0%	7	19.4%	22	17.9%	34	14.9%
	church	6	5.8%	19	9.9%	0	0.0%	3	8.3%	6	4.9%	22	9.6%
	gas station	6	5.8%	7	3.6%	2	10.0%	2	5.6%	8	6.5%	9	3.9%
	bar	8	7.8%	26	13.5%	0	0.0%	6	16.7%	8	6.5%	32	14.0%
	lime	16	15.5%	38	19.8%	2	10.0%	13	36.1%	18	14.6%	51	22.4%
	gym	2	1.9%	5	2.6%	0	0.0%	0	0.0%	2	1.6%	5	2.2%
	school	23	22.3%	37	19.3%	4	20.0%	3	8.3%	27	22.0%	40	17.5%
	motel	1	1.0%	3	1.6%	2	10.0%	2	5.6%	3	2.4%	5	2.2%
	other	4	3.9%	5	2.6%	1	5.0%	2	5.6%	5	4.1%	7	3.1%
Age at sexual debut													
	<=10	15	14.6%	21	10.9%	0	0.0%	4	11.1%	15	12.2%	25	11.0%
	11-15	65	63.1%	98	51.0%	12	60.0%	20	55.6%	77	62.6%	118	51.8%
	16-17	20	19.4%	48	25.0%	8	40.0%	6	16.7%	28	22.8%	54	23.7%
	18+	2	1.9%	24	12.5%	0	0.0%	6	16.7%	2	1.6%	30	13.2%
	Missing	1	1.0%	1	0.5%	0	0.0%	0	0.0%	1	0.8%	1	0.4%

Condom use at last sexwith most recent male partner													
	Used condom	67	71.3%	122	68.9%	11	61.1%	21	58.3%	78	69.6%	143	67.1%
	Did not use condom	27	28.7%	55	31.1%	7	38.9%	15	41.7%	34	30.4%	70	32.9%
Condom use at last sex with most recent partner - any sex													
	Used condom	67	67.7%	123	68.3%	11	61.1%	22	61.1%	78	66.7%	145	67.1%
	Did not use condom	32	32.3%	57	31.7%	7	38.9%	14	38.9%	39	33.3%	71	32.9%
Condom use with most recent regular partner													
	Used condom	42	55.3%	74	56.9%	10	58.8%	12	52.2%	52	55.9%	86	56.2%
	Did not use condom	34	44.7%	56	43.1%	7	41.2%	11	47.8%	41	44.1%	67	43.8%
In the last 6 months when you had sexual intercourse, did you or your partner ever put the condom on after you already started having sex?													
	Yes	17	16.5%	32	16.7%	3	15.0%	9	25.0%	20	16.3%	41	18.0%
	No	86	83.5%	159	82.8%	17	85.0%	26	72.2%	103	83.7%	185	81.1%
	Don't know	0	0.0%	1	0.5%	0	0.0%	1	2.8%	0	0.0%	2	0.9%
In the last 6 months when you had sexual intercourse, did you or your partner ever take the condom off before you were finished having sex?													
	Yes	25	24.3%	52	27.1%	4	20.0%	16	44.4%	29	23.6%	68	29.8%
	No	78	75.7%	140	72.9%	16	80.0%	20	55.6%	94	76.4%	160	70.2%
In the last 6 months when you had sexual intercourse, did the condom ever break/leak during sex or while pulling out?													
	Yes	34	33.0%	56	29.2%	7	35.0%	12	33.3%	41	33.3%	68	29.8%
	No	67	65.0%	132	16.7%	13	65.0%	23	63.9%	80	65.0%	155	68.0%
	Don't know	2	1.9%	4	16.7%	0	0.0%	1	2.8%	2	1.6%	5	2.2%
In the last 6 months when you had sexual intercourse, did the condom ever slip off during sex or while pulling out?													
	Yes	23	22.3%	33	16.7%	4	20.0%	8	22.2%	27	22.0%	41	18.0%
	No	80	77.7%	156	16.7%	16	80.0%	28	77.8%	96	78.0%	184	80.7%
	Don't know	0	0.0%	1	16.7%	0	0.0%	0	0.0%	0	0.0%	1	0.4%

Table 5a: Condom Affordability, Accessibility, and Lubricant Use

		MSM				TGW				Total			
		16-19 (n=103)		20-24 (n=192)		16-19 (n=20)		20-24 (n=36)		16-19 (n=123)		20-24 (n=228)	
		n	%	n	%	n	%	n	%	n	%	n	%
Condom Affordability													
	Very affordable	65	63.1%	133	69.3%	10	50.0%	21	58.3%	75	61.0%	154	67.5%
	Somewhat affordable	19	18.4%	44	22.9%	7	35.0%	10	27.8%	26	21.1%	54	23.7%
	Not affordable	12	11.7%	10	5.2%	1	5.0%	4	11.1%	13	10.6%	14	6.1%
	Don't Know	7	6.8%	5	2.6%	2	10.0%	1	2.8%	9	7.3%	6	2.6%
Condom Accessibility													
	Easy to access	61	59.2%	135	70.3%	12	60.0%	24	66.7%	73	59.3%	159	69.7%
	Somewhat easy to access	34	33.0%	510	265.6%	7	35.0%	9	25.0%	41	33.3%	519	227.6%
	Not easy to access	5	4.9%	5	2.6%	1	5.0%	1	2.8%	6	4.9%	6	2.6%
	Do not know	3	2.9%	1	0.5%	0	0.0%	2	5.6%	3	2.4%	3	1.3%
	Missing	0	0.0%	1	0.5%	0	0.0%	0	0.0%	0	0.0%	1	0.4%
Received Condom for Free													
	Yes	68	66.0%	149	77.6%	13	65.0%	29	80.6%	81	65.9%	178	78.1%
	No	35	34.0%	42	21.9%	7	35.0%	7	19.4%	42	34.1%	49	21.5%
	Don't know	0	0.0%	1	0.5%	0	0.0%	0	0.0%	0	0.0%	1	0.4%
Ever Used Female Condom													
	Used female condom	3	2.9%	3	1.6%	0	0.0%	3	8.3%	3	2.4%	6	2.6%
	Did not use	100	97.1%	188	97.9%	20	100.0%	33	91.7%	120	97.6%	221	96.9%
	Missing	0	0.0%	1	0.5%	0	0.0%	0	0.0%	0	0.0%	1	0.4%
Condom Use* at Last Sex Act													
	Yes	67	65.0%	123	64.1%	11	55.0%	22	61.1%	78	63.4%	145	63.6%
	No	32	31.1%	57	29.7%	7	35.0%	14	38.9%	39	31.7%	71	31.1%
	Missing	4	3.9%	12	6.3%	2	10.0%	0	0.0%	6	4.9%	12	5.3%
Frequency of Lubricant Use													
	Always	34	33.0%	77	40.1%	9	45.0%	20	55.6%	43	35.0%	97	42.5%
	Usually	21	20.4%	31	16.1%	3	15.0%	4	11.1%	24	19.5%	35	15.4%
	Sometimes	34	33.0%	51	26.6%	6	30.0%	6	16.7%	40	32.5%	57	25.0%
	Rarely	7	6.8%	17	8.9%	2	10.0%	3	8.3%	9	7.3%	20	8.8%
	Never	7	6.8%	16	8.3%	0	0.0%	3	8.3%	7	5.7%	19	8.3%

Table 6a: Drug and Alcohol Use of MSM and TGW Among Adolescents

			MSM				TGW				Total			
			16-19 (n=103)		20-24 (n=192)		16-19 (n=20)		20-24 (n=36)		16-19 (n=123)		20-24 (n=228)	
			n	%	n	%	n	%	n	%	n	%	n	%
How often has 6 or more drinks														
	Never		66	64.1%	107	55.7%	14	70.0%	22	61.1%	80	65.0%	129	56.6%
	Less than once a month		19	18.4%	49	25.5%	3	15.0%	7	19.4%	22	17.9%	56	24.6%
	Monthly		12	11.7%	13	6.8%	0	0.0%	1	2.8%	12	9.8%	14	6.1%
	Weekly		4	3.9%	13	6.8%	2	10.0%	1	2.8%	6	4.9%	14	6.1%
	Daily		2	1.9%	9	4.7%	0	0.0%	5	13.9%	2	1.6%	14	6.1%
	Missing			0.0%	1	0.5%	1	5.0%	0	0.0%	1	0.8%	1	0.4%
Audit-C Designation														
	Hazardous drinker		25	24.3%	65	33.9%	5	25.0%	11	30.6%	30	24.4%	76	33.3%
	Not a hazardous drinker		78	75.7%	127	66.1%	15	75.0%	25	69.4%	93	75.6%	152	66.7%
Drug Use Frequency														
	Heroin													
		Did not use	103	100.0%	190	99.0%	20	100.0%	36	100.0%	123	100.0%	226	99.1%
		Used at least once in past month	0	0.0%	1	0.5%	0	0.0%	0	0.0%	0	0.0%	1	0.4%
		Declined/missing		0.0%		0.0%		0.0%		0.0%	0	0.0%	0	0.0%
	Cocaine													
		Did not use	100	97.1%	190	99.0%	19	95.0%	36	100.0%	119	96.7%	226	99.1%
		Used at least once in past month	2	1.9%	2	1.0%	1	5.0%	0	0.0%	3	2.4%	2	0.9%
		Declined/missing	1	1.0%		0.0%		0.0%		0.0%	1	0.8%	0	0.0%
	Crack													
		Did not use	102	99.0%	191	99.5%	20	100.0%	36	100.0%	122	99.2%	227	99.6%
		Used at least once in past month	0	0.0%	1	0.5%	0	0.0%	0	0.0%	0	0.0%	1	0.4%
		Declined/missing	1	1.0%		0.0%		0.0%		0.0%	1	0.8%	0	0.0%
	Meth													
		Did not use	103	100.0%	192	100.0%	20	100.0%	36	100.0%	123	100.0%	228	100.0%
		Used at least once in past month	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
		Declined/missing		0.0%		0.0%		0.0%		0.0%	0	0.0%	0	0.0%
	Speed													
		Did not use	103	100.0%	189	98.4%	20	100.0%	36	100.0%	123	100.0%	225	98.7%
		Used at least once in past month	0	0.0%	2	1.0%	0	0.0%	0	0.0%	0	0.0%	2	0.9%
		Declined/missing		0.0%	1	0.5%		0.0%		0.0%	0	0.0%	1	0.4%
	Other opioids													
		Did not use	102	99.0%	189	98.4%	20	100.0%	35	97.2%	122	99.2%	224	98.2%
		Used at least once in past month	1	1.0%	1	0.5%	0	0.0%	1	2.8%	1	0.8%	2	0.9%
		Declined/missing		0.0%		0.0%		0.0%	1	2.8%	0	0.0%	1	0.4%
	Solvents													
		Did not use	98	95.1%	184	95.8%	20	100.0%	36	100.0%	118	95.9%	220	96.5%
		Used at least once in past month	3	2.9%	6	3.1%	0	0.0%	0	0.0%	3	2.4%	6	2.6%
		Declined/missing	2	1.9%	2	1.0%		0.0%		0.0%	2	1.6%	2	0.9%
	Hallucinogens													
		Did not use	101	98.1%	185	96.4%	20	100.0%	36	100.0%	121	98.4%	221	96.9%
		Used at least once in past month	2	1.9%	6	3.1%	0	0.0%	0	0.0%	2	1.6%	6	2.6%
		Declined/missing		0.0%	1	0.5%		0.0%		0.0%	0	0.0%	1	0.4%
	Marijuana													

		Did not use	66	64.1%	101	52.6%	11	55.0%	18	50.0%	77	62.6%	119	52.2%
		monthly or less	9	8.7%	16	8.3%	4	20.0%	2	5.6%	13	10.6%	18	7.9%
		Several times a month	5	4.9%	19	9.9%	1	5.0%	1	2.8%	6	4.9%	20	8.8%
		2-4 times per month	5	4.9%	15	7.8%	2	10.0%	3	8.3%	7	5.7%	18	7.9%
		2-3 times per week	4	3.9%	8	4.2%	1	5.0%	3	8.3%	5	4.1%	11	4.8%
		4 or more times per week	14	13.6%	32	16.7%	1	5.0%	9	25.0%	15	12.2%	41	18.0%
		Declined/missing	0	0.0%	1	0.5%	0	0.0%	0	0.0%	0	0.0%	1	0.4%
Ever injected drugs														
	Yes		0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
	No		103	100.0%	192	100.0%	23	115.0%	38	105.6%	126	102.4%	230	100.9%

Table 7a: STI Diagnosis, Symptoms, and Treatment of MSM and TGW Among Adolescents

		MSM				TGW				Total			
		16-19 (n=103)		20-24 (n=192)		16-19 (n=20)		20-24 (n=36)		16-19 (n=123)		20-24 (n=228)	
		n	%	n	%	n	%	n	%	n	%	n	%
Clinician diagnosed STI in previous 12 months													
	Yes	24	23.3%	47	24.5%	4	17.4%	12	31.6%	28	22.2%	59	25.7%
	No	78	75.7%	145	75.5%	16	69.6%	23	60.5%	94	74.6%	168	73.0%
	Don't know/missing	1	1.0%	0	0.0%	0	0.0%	1	2.6%	1	0.8%	1	0.4%
STI symptomology in previous 12 months													
	Yes	21	20.4%	45	23.4%	6	26.1%	7	18.4%	27	21.4%	52	22.6%
	No	79	76.7%	146	76.0%	14	60.9%	28	73.7%	93	73.8%	174	75.7%
	Don't know/missing	1	1.0%	0	0.0%	0	0.0%	1	2.6%	1	0.8%	1	0.4%
Sought treatment for STI (out of those with symptoms or diagnosis)													
	Yes	23	82.1%	56	86.2%	4	66.7%	9	75.0%	27	79.4%	65	84.4%
	No	5	17.9%	9	13.8%	2	33.3%	3	25.0%	7	20.6%	12	15.6%

Table 8a: Stigma, Discrimination, and Incarceration in MSM and TGW Among Adolescents													
		MSM				TGW				Total			
		16-19 (n=103)		20-24 (n=192)		16-19 (n=20)		20-24 (n=36)		16-19 (n=123)		20-24 (n=228)	
		n	%	n	%	n	%	n	%	n	%	n	%
Refused services in previous 12 months due to MSM/trans status													
	Healthcare	0	0.0%	3	1.6%	1	5.0%	4	11.1%	1	0.8%	7	3.1%
	Employment	9	8.7%	12	6.3%	1	5.0%	13	36.1%	10	8.1%	25	11.0%
	Religious services	8	7.8%	11	5.7%	2	10.0%	8	22.2%	10	8.1%	19	8.3%
	Service (i.e. restaurant)	5	4.9%	7	3.6%	1	5.0%	10	27.8%	6	4.9%	17	7.5%
	Housing	12	11.7%	22	11.5%	3	15.0%	13	36.1%	15	12.2%	35	15.4%
	Police assistance	10	9.7%	31	16.1%	4	20.0%	15	41.7%	14	11.4%	46	20.2%
Experienced verbal insults in previous 12 months													
	Yes	77	74.8%	123	64.1%	17	85.0%	28	77.8%	94	76.4%	151	66.2%
	No	26	25.2%	68	35.4%	3	15.0%	8	22.2%	29	23.6%	76	33.3%
	Don't know/missing	0	0.0%	1	0.5%	0	0.0%	0	0.0%	0	0.0%	1	0.4%
Experienced physical violence in previous 12 months													
	Yes	18	17.5%	19	9.9%	6	30.0%	16	44.4%	24	19.5%	35	15.4%
	No	85	82.5%	172	89.6%	14	70.0%	20	55.6%	99	80.5%	192	84.2%
	Don't know/missing	0	0.0%	1	0.5%	0	0.0%	0	0.0%	0	0.0%	1	0.4%
Was forced into sex in previous 12 months													
	Yes	7	6.8%	12	6.3%	3	15.0%	5	13.9%	10	8.1%	17	7.5%
	No	96	93.2%	180	93.8%	17	85.0%	31	86.1%	113	91.9%	211	92.5%
	Don't know/missing	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Has ever been to jail/prison													
	Yes	20	19.4%	46	24.0%	4	20.0%	15	41.7%	24	19.5%	61	26.8%
	No	83	80.6%	154	80.2%	16	80.0%	21	58.3%	99	80.5%	175	76.8%
	Refused	0	0.0%	1	0.5%	0	0.0%	0	0.0%	0	0.0%	1	0.4%

Table 9a: Mental Health Among Adolescent MSM and TGW													
		MSM				TGW				Total			
		16-19 (n=103)		20-24 (n=192)		16-19 (n=20)		20-24 (n=36)		16-19 (n=123)		20-24 (n=228)	
		n	%	n	%	n	%	n	%	n	%	n	%
Anxiety measure from PHQ-4													
	Anxiety suggested	28	27.2%	41	21.4%	8	40.0%	9	25.0%	36	29.3%	50	21.9%
	No anxiety suggested	74	71.8%	150	78.1%	12	60.0%	27	75.0%	86	69.9%	177	77.6%
	Missing	1	1.0%	1	0.5%		0.0%		0.0%	1	0.8%	1	0.4%
Depression measure from PHQ-4													
	Depression suggested	16	15.5%	33	17.2%	3	15.0%	10	27.8%	19	15.4%	43	18.9%
	No depression suggested	87	84.5%	159	82.8%	17	85.0%	26	72.2%	104	84.6%	185	81.1%
	Missing	0	0.0%	0	0	0	0.0%	0	0.0%	0	0.0%	0	0.0%
Combined PHQ-4 anxiety and depression score													
	Normal	51	49.5%	89	46.4%	7	35.0%	12	33.3%	58	47.2%	101	44.3%
	Mild	33	32.0%	74	38.5%	8	40.0%	17	47.2%	41	33.3%	91	39.9%
	Moderate	14	13.6%	22	11.5%	5	25.0%	5	13.9%	19	15.4%	27	11.8%
	Severe	4	3.9%	6	3.1%	0	0.0%	2	5.6%	4	3.3%	8	3.5%
	Missing	1	1.0%	1	0.5%	0	0.0%	0	0.0%	1	0.8%	1	0.4%
Food insecurity: Skipped or reduced meals due to lack of money													
	Yes	22	21.4%	59	30.7%	5	25.0%	13	36.1%	27	22.0%	72	31.6%
	No	81	78.6%	133	69.3%	15	75.0%	23	63.9%	96	78.0%	156	68.4%

Table 10a: Service Provision Among Adolescent MSM and TGW													
		MSM				TGW				Total			
		16-19 (n=103)		20-24 (n=192)		16-19 (n=20)		20-24 (n=36)		16-19 (n=123)		20-24 (n=228)	
		n	%	n	%	n	%	n	%	n	%	n	%
Received any information about how to prevent transmission of HIV from someone at the Ministry of Health or any other health service organization in previous 12 months													
	Yes	80	77.7%	146	76.0%	12	60.0%	28	77.8%	92	74.8%	174	76.3%
	No	23	22.3%	46	24.0%	6	30.0%	8	22.2%	29	23.6%	54	23.7%
	Don't know/missing	0	0.0%	0	0.0%	2	10.0%	0	0.0%	2	1.6%	0	0.0%
Aware of any civil society or religious organization(s) that deliver non-medical assistance or advice to people who are MSM/Trans													
	Yes	40	38.8%	83	43.2%	9	45.0%	16	44.4%	49	39.8%	99	43.4%
	No	63	61.2%	107	55.7%	10	50.0%	20	55.6%	73	59.3%	127	55.7%
	Don't know/missing	0	0.0%	2	1.0%	1	5.0%	0	0.0%	1	0.8%	2	0.9%
Attended any meetings or programs to discuss HIV and/or AIDS in previous 6 months													
	Yes	46	44.7%	100	52.1%	11	55.0%	26	72.2%	57	46.3%	126	55.3%
	No	57	55.3%	90	46.9%	9	45.0%	10	27.8%	66	53.7%	100	43.9%
	Don't know/missing	0	0.0%	2	1.0%	0	0.0%	0	0.0%	0	0.0%	2	0.9%
Contact with a health peer educator in the community in previous 6 months													
	Yes	25	24.3%	71	37.0%	5	25.0%	12	33.3%	30	24.4%	83	36.4%
	No	78	75.7%	120	62.5%	14	70.0%	24	66.7%	92	74.8%	144	63.2%
	Don't know/missing	0	0.0%	1	0.5%	1	5.0%	0	0.0%	1	0.8%	1	0.4%

Table 13a: HIV Testing Behavior Among Adolescent MSM and TGW													
		MSM				TGW				Total			
		16-19 (n=103)		20-24 (n=192)		16-19 (n=20)		20-24 (n=36)		16-19 (n=123)		20-24 (n=228)	
		n	%	n	%	n	%	n	%	n	%	n	%
Previous tested for HIV													
	Yes	93	90.3%	185	96.4%	19	95.0%	36	100.0%	112	91.1%	221	96.9%
	No	10	9.7%	7	3.6%	1	5.0%	0	0.0%	11	8.9%	7	3.1%
Used HIV self-testing kit													
	Yes	14	13.6%	36	18.8%	0	0.0%	4	11.1%	14	11.4%	40	17.5%
	No	89	86.4%	156	81.3%	20	100.0%	32	88.9%	109	88.6%	188	82.5%
Willingness to use HIV self-test in the future													
	Very unlikely	11	10.7%	31	16.1%	3	15.0%	8	22.2%	14	11.4%	39	17.1%
	Somewhat unlikely	8	7.8%	9	4.7%	2	10.0%	0	0.0%	10	8.1%	9	3.9%
	Neutral	17	16.5%	26	13.5%	3	15.0%	5	13.9%	20	16.3%	31	13.6%
	Somewhat likely	18	17.5%	38	19.8%	2	10.0%	8	22.2%	20	16.3%	46	20.2%
	Very likely	47	45.6%	86	44.8%	9	45.0%	15	41.7%	56	45.5%	101	44.3%
	Don't know/missing	2	1.9%	2	1.0%	1	5.0%	0	0.0%	3	2.4%	2	0.9%
Time since last HIV test													
	0-3 months	52	50.5%	86	44.8%	8	40.0%	9	25.0%	60	48.8%	95	41.7%
	4-6 months	20	19.4%	39	20.3%	5	25.0%	13	36.1%	25	20.3%	52	22.8%
	7 months - 1 year	17	16.5%	23	12.0%	3	15.0%	11	30.6%	20	16.3%	34	14.9%
	Greater than 1 year	4	3.9%	36	18.8%	3	15.0%	3	8.3%	7	5.7%	39	17.1%
	Never tested	10	9.7%	8	4.2%	1	5.0%	0	0.0%	11	8.9%	8	3.5%

Table 14a: PrEP Knowledge and Use Among HIV-Negative and Unaware HIV-Positive Adolescent MSM and TGW													
		MSM				TGW				Total			
		16-19 (n=89)		20-24 (n=148)		16-19 (n=13)		20-24 (n=21)		16-19 (n=102)		20-24 (n=169)	
		n	%	n	%	n	%	n	%	n	%	n	%
Aware of PrEP													
	Yes	38	43.2%	70	47.3%	4	30.8%	8	38.1%	42	41.2%	78	46.2%
	No	50	56.8%	76	51.4%	9	69.2%	12	57.1%	59	57.8%	88	52.1%
	Missing	1	1.1%	2	1.4%	0	0.0%	1	4.8%	1	1.0%	3	1.8%
Willingness to Use PrEP													
	Very willing	58	65.9%	86	58.1%	8	61.5%	16	76.2%	66	64.7%	102	60.4%
	Moderately willing	10	11.4%	18	12.2%	2	15.4%	1	4.8%	12	11.8%	19	11.2%
	Somewhat willing	8	9.1%	22	14.9%	2	15.4%	1	4.8%	10	9.8%	23	13.6%
	Not willing	13	14.8%	22	14.9%	1	7.7%	3	14.3%	14	13.7%	25	14.8%

Table 15a: Syphilis Test Results and HIV Prevalence Among Adolescent MSM and TGW

		MSM				TGW				Total			
		16-19 (n=103)		20-24 (n=192)		16-19 (n=20)		20-24 (n=36)		16-19 (n=123)		20-24 (n=228)	
		n	%	n	%	n	%	n	%	n	%	n	%
Syphilis Test Result													
	Positive	19	18.4%	63	32.8%	7	35.0%	13	36.1%	26	21.1%	76	33.3%
	Negative	84	81.6%	129	67.2%	13	65.0%	23	63.9%	97	78.9%	152	66.7%
HIV rapid test results													
	HIV-Positive	11	10.7%	45	23.4%	7	35.0%	15	41.7%	18	14.6%	60	26.3%
	HIV-Negative	92	89.3%	147	76.6%	13	65.0%	21	58.3%	105	85.4%	168	73.7%
Previously known HIV infection (out of HIV+)													
	Newly diagnosed	2	18.2%	20	44.4%	5	71.4%	5	33.3%	7	5.7%	25	11.0%
	Previously known HIV+	9	81.8%	25	55.6%	2	28.6%	10	66.7%	11	8.9%	35	15.4%
ART use among previously known HIV+		SAMPLE SIZES TOO SMALL FOR MEANINGFUL INTERPRETATION											
	Currently on ART												
	Not on ART												
Result among those on ART													
	Virally suppressed												
	Not virally suppressed												

Appendix B: Selected Tables by HIV Sample Prevalence

Table 2b: Sample Prevalence of HIV by Demographics Characteristics							
		MSM (n=532)		TGW (n=102)		Total (n=648)	
		Crude		Crude		Crude	
		# HIV+	%	# HIV+	%	# HIV+	%
Sex at Birth							
	Male	158	29.8%	52	51.0%	213	33.3%
	Female	0	0.0%	0	0.0%	0	0.0%
Current Gender							
	Male	132	28.4%	0	0.0%	132	28.4%
	Female-to-male	0	0.0%	0	0.0%	2	15.4%
	Male-to-female	0	0.0%	52	51.0%	52	51.0%
	Genderqueer	23	37.1%	0	0.0%	24	38.1%
	Other	3	50.0%	0	0.0%	3	100.0%
Age							
	16-17	3	10.3%	3	60.0%	6	17.6%
	18-24	53	20.2%	19	37.3%	72	22.6%
	25+	102	42.5%	30	65.2%	135	45.8%
Age							
	16-17	3	10.3%	3	60.0%	6	17.6%
	18-24	53	20.2%	19	37.3%	72	22.6%
	25-29	46	36.2%	17	56.7%	64	40.0%
	30-39	38	50.7%	13	81.3%	52	54.7%
	40-49	14	48.3%	0	0.0%	15	48.4%
	50+	4	44.4%	0	0.0%	4	44.4%
Sexual Orientation							
	Gay/Lesbian	90	36.3%	38	48.1%	130	38.8%
	Bisexual	64	23.3%	7	53.8%	72	24.5%
	Heterosexual/Straight	4	50.0%	7	70.0%	11	61.1%
Highest education level completed							
	Primary or less	3	37.5%	2	100.0%	5	50.0%
	Secondary	99	29.9%	32	55.2%	133	33.3%
	University or higher	15	22.4%	3	37.5%	18	23.7%
	Vocational	40	32.3%	15	44.1%	56	34.8%
	Other	1	50.0%	0	0.0%	1	50.0%
Current student status							
	Full-time student	15	16.5%	5	35.7%	20	18.9%
	Part-time student	9	40.9%	4	57.1%	13	33.3%
	Not a student	134	32.7%	43	53.1%	180	35.8%
Household Monthly Income							
	<=40,000 JMD	74	35.6%	25	53.2%	99	37.9%
	41k - 149k JMD	49	29.2%	10	41.7%	59	30.4%
	150k - 299k JMD	8	16.7%	2	50.0%	10	19.2%
	300k - 499k	3	20.0%	1	50.0%	4	21.1%
	>=450k	1	9.1%	2	100.0%	3	23.1%
	Do not know	20	30.3%	11	61.1%	33	38.4%
	Declined	3	20.0%	1	20.0%	4	19.0%
Occupation							
	Not employed	60	30.3%	23	47.9%	83	33.2%
	Professional	28	24.8%	2	25.0%	30	24.4%
	Laborer(factory/construction)	15	36.6%	3	37.5%	19	38.0%
	Agriculture/fishing	2	40.0%	2	66.7%	5	55.6%
	Driver	2	28.6%	1	50.0%	3	33.3%
	Retail/restaurant/performer	24	31.2%	6	46.2%	30	31.9%
	Security/military/police	7	25.9%	1	33.3%	8	26.7%
	Sex work	5	41.7%	10	100.0%	15	68.2%
	Call center	12	27.9%	2	66.7%	14	29.8%

	Other	4	44.4%	2	50.0%	6	46.2%
Parish of Residence							
	Clarendon	14	40.0%	6	37.5%	20	39.2%
	Hanover	4	30.8%	0	0.0%	4	28.6%
	Kingston	45	29.4%	19	50.0%	64	32.7%
	Manchester	0	0.0%	2	100.0%	2	40.0%
	Portland	1	20.0%	0	0.0%	1	16.7%
	St. Andrew	13	28.3%	2	66.7%	15	29.4%
	St. Ann	15	45.5%	9	60.0%	24	50.0%
	St. Catherine	18	25.7%	4	30.8%	24	27.3%
	St. Elizabeth	1	33.3%	0	0.0%	1	33.3%
	St. James	28	27.2%	2	66.7%	30	28.3%
	St. Mary	5	55.6%	1	33.3%	6	50.0%
	St. Thomas	1	7.7%	0	0.0%	2	14.3%
	Trelawney	1	33.3%	4	100.0%	5	71.4%
	Westmoreland	12	27.9%	3	75.0%	15	31.9%

Table 4b: Sample Prevalence of HIV by Sexual Behavior

		MSM (n=532)		TGW (n=102)		Total (n=648)	
		Crude		Crude		Crude	
		# HIV+	%	# HIV+	%	# HIV+	%
Total number of male partners in previous 12 months							
	0	1	12.5%	0	0.0%	1	7.7%
	1	36	30.3%	11	45.8%	47	32.2%
	2 or more	122	30.0%	41	52.6%	165	33.7%
Total number of female partners in previous 12 months							
	0	113	34.5%	40	47.6%	155	37.4%
	1	18	26.5%	4	57.1%	22	28.2%
	2 or more	28	20.7%	8	72.7%	36	23.4%
Paid for sex with any partners in previous 12 months							
	Yes	21	32.3%	12	60.0%	33	37.9%
	No	137	29.3%	40	48.8%	180	32.1%
Received payment for sex with any partners in previous 12 months							
	Yes	44	32.8%	32	58.2%	77	39.5%
	No	114	28.6%	20	42.6%	136	30.0%
Sexual position preference							
	Receptive	38	28.1%	24	47.1%	63	33.5%
	Insertive	30	22.4%	6	60.0%	37	25.5%
	Oral Only	1	12.5%	0	0.0%	1	8.3%
	Versatile	89	35.2%	22	53.7%	112	37.3%
	None	0	0.0%	0		0	0.0%
Age at sexual debut							
	10 and under	23	33.8%	7	58.3%	31	37.8%
	11-15	72	27.1%	26	53.1%	99	30.9%
	16-17	34	30.9%	11	42.3%	45	32.8%
	18+	27	32.9%	8	53.3%	36	36.7%
	Missing						
Venues for meeting sexual partners							
	hotel	17	43.6%	7	58.3%	24	44.4%
	street	81	31.8%	37	56.9%	121	36.7%
	friends	93	30.8%	31	50.8%	125	33.8%
	internet	128	30.8%	40	48.2%	170	33.4%
	party	54	34.2%	27	50.9%	83	37.9%
	work	41	30.4%	15	53.6%	57	34.8%
	workshop	24	30.4%	13	56.5%	37	36.3%
	church	22	34.9%	7	58.3%	29	38.7%
	gas station	14	41.2%	7	58.3%	21	45.7%
	bar	22	29.7%	11	55.0%	34	33.7%
	lime	29	29.6%	13	48.1%	42	33.3%
	gym	5	41.7%	0	0.0%	5	38.5%
	school	21	21.9%	5	45.5%	26	23.9%
	motel	5	50.0%	8	100.0%	13	72.2%
	other	9	39.1%	4	50.0%	13	40.6%

Table 5b: HIV Sample Prevalence by Condom Use at Last Sex among MSM and TGP

		MSM (n=537)		TGW (n=102)		Total (n=652)	
		Crude		Crude		Crude	
		# HIV+	%	# HIV+	%	# HIV+	%
Condom Use* at Last Sex Act							
	Yes	119	33.9%	32	54.2%	151	36.8%
	No	33	20.8%	17	44.7%	50	25.4%

Table 6b: Sample Prevalence of HIV by Drug and Alcohol Use

			MSM (n=532)		TGW (n=102)		Total (n=648)	
			Crude		Crude		Crude	
			# HIV+	%	# HIV+	%	# HIV+	%
How often has 6 or more drinks								
	Never		105	33.8%	36	54.5%	143	37.2%
	Less than once a month		30	25.9%	9	52.9%	39	28.9%
	Monthly		10	20.4%	2	50.0%	12	21.4%
	Weekly		12	27.9%	1	16.7%	13	26.0%
	Daily		2	15.4%	3	37.5%	5	23.8%
Audit-C Designation								
	Hazardous drinker		45	26.2%	12	41.4%	57	27.9%
	Not a hazardous drinker		114	31.6%	40	54.8%	156	35.1%
Drug Use Frequency								
	Marijuana							
		Did not use	90	28.4%	21	43.8%	113	30.6%
		Used at least once	69	32.1%	31	57.4%	100	36.0%

Table 7b: STI Diagnosis, Symptomology, and Treatment

		MSM (n=532)		TGW (n=102)		Total (n=648)	
		Crude		Crude		Crude	
		# HIV+	%	# HIV+	%	# HIV+	%
Clinician diagnosed STI in previous 12 months							
	Yes	67	48.9%	20	64.5%	88	52.1%
	No	92	23.3%	31	44.3%	124	26.1%
STI symptomology in previous 12 months							
	Yes	52	45.2%	20	64.5%	73	49.7%
	No	107	25.8%	32	45.7%	140	28.3%
Sought treatment for STI (out of those with symptoms or diagnosis)							
	Yes	71	48.0%	21	65.6%	93	51.4%
	No	15	60.0%	7	58.3%	22	59.5%
Syphilis/HIV Co-infection							
	Syphilis positive	93	56.7%	29	65.9%	124	59.0%
	Syphilis negative	66	17.9%	23	39.7%	89	20.3%

Table 8b: HIV Sample Prevalence by Stigma, Discrimination, and Incarceration among MSM and TGP

		MSM (n=532)		TGW (n=102)		Total (n=648)	
		Crude		Crude		Crude	
		# HIV+	%	# HIV+	%	# HIV+	%
Refused services in previous 12 months due to MSM/trans status							
	Healthcare	10	83.3%	6	60.0%	16	69.6%
	Employment	12	26.7%	17	60.7%	29	37.7%
	Religious services	19	48.7%	10	43.5%	29	46.0%
	Service (i.e. restaurant)	15	53.6%	13	46.4%	28	50.0%
	Housing	23	35.4%	15	46.9%	38	38.0%
	Police assistance	37	44.0%	26	59.1%	65	48.9%
Experienced verbal insults in previous 12 months							
	Yes	96	27.9%	43	50.6%	141	32.0%
	No	63	33.7%	9	52.9%	72	35.1%
Experienced physical violence in previous 12 months							
	Yes	28	43.8%	26	55.3%	54	48.2%
	No	130	27.8%	26	47.3%	158	29.6%
Was forced into sex in previous 12 months							
	Yes	7	24.1%	13	72.2%	20	40.8%
	No	152	30.2%	39	46.4%	193	32.2%
Has ever been to jail/prison							
	Yes	63	40.9%	28	65.1%	93	45.4%
	No	95	25.1%	24	40.7%	119	26.9%

Table 9b: HIV Sample Prevalence by Mental Health Among MSM and TGP

		MSM (n=532)		TGW (n=102)		Total (n=648)	
		Crude		Crude		Crude	
		# HIV+	%	# HIV+	%	# HIV+	%
Anxiety measure from PHQ-4							
	Anxiety suggested	32	33.0%	14	43.8%	47	35.3%
	No anxiety suggested	126	29.0%	38	54.3%	165	32.2%
Depression measure from PHQ-4							
	Depression suggested	28	35.9%	17	63.0%	46	41.8%
	No depression suggested	130	28.6%	35	46.7%	166	30.9%
Combined PHQ-4 anxiety and depression score							
	Normal	76	27.0%	19	63.3%	95	30.4%
	Mild	56	31.1%	21	43.8%	78	33.1%
	Moderate	23	39.7%	10	50.0%	34	42.0%
	Severe	2	18.2%	2	50.0%	4	26.7%
Self-harm thoughts in previous 12 months							
	Yes	51	32.7%	23	46.0%	75	35.5%
	No	108	28.6%	29	55.8%	138	31.6%
Self-harm thoughts in previous 4 weeks							
	Yes	36	35.0%	14	41.2%	50	36.0%
	No	123	28.6%	38	55.9%	163	32.0%
Attempted self harm							
	Yes	16	34.0%	11	50.0%	27	38.0%
	No	143	29.4%	41	51.3%	186	32.2%
Food insecurity: Skipped or reduced meals due to lack of money							
	Yes	58	36.9%	22	61.1%	80	41.0%
	No	101	26.9%	30	45.5%	133	29.4%

Table 10b: HIV Sample Prevalence by Service Provision among MSM and TGP							
		MSM (n=532)		TGW (n=102)		Total (n=648)	
		Crude		Crude		Crude	
		# HIV+	%	# HIV+	%	# HIV+	%
Received any information about how to prevent transmission of HIV from someone at the Ministry of Health or any other health service organization in previous 12 months							
	Yes	121	30.3%	40	53.3%	161	33.6%
	No	38	28.6%	11	44.0%	51	30.5%
Aware of any civil society or religious organization(s) that deliver non-medical assistance or advice to people who are MSM/Trans							
	Yes	62	29.2%	30	51.7%	93	33.8%
	No	96	30.4%	21	48.8%	118	32.2%
Attended any meetings or programs to discuss HIV and/or AIDS in previous 6 months							
	Yes	83	31.8%	34	53.1%	117	35.7%
	No	76	28.1%	18	47.4%	96	30.2%
Contact with a health peer educator in the community in previous 6 months							
	Yes	62	32.3%	17	48.6%	79	34.5%
	No	97	28.5%	34	51.5%	133	31.9%

Table 13b: HIV Sample Prevalence by HIV Testing Behavior among MSM and TGP

		MSM (n=536)		TGW (n=102)		Total (n=652)	
		Crude		Crude		Crude	
		# HIV+	%	# HIV+	%	# HIV+	%
Previous tested for HIV							
	Yes	151	29.8%	50	50.5%	203	33.0%
	No	7	26.9%	2	66.7%	9	29.0%
Heard of HIV self testing							
	Yes	28	26.7%	4	57.1%	33	28.7%
	No	131	30.6%	48	50.5%	180	33.8%
Used HIV self testing							
	Yes	2	18.2%	1	100.0%	3	23.1%
	No	26	27.7%	3	50.0%	30	29.4%
Willingness to use HIV self-test in the future							
	Very unlikely	24	32.0%	11	64.7%	35	37.2%
	Somewhat unlikely	9	33.3%	1	33.3%	10	32.3%
	Neutral	32	40.5%	10	55.6%	44	43.6%
	Somewhat likely	30	29.7%	13	59.1%	43	34.7%
	Very likely	59	24.2%	15	37.5%	74	25.6%
Time since last HIV test							
	0-3 months	10	5.1%	3	15.0%	13	6.0%
	4-6 months	7	9.2%	5	23.8%	12	12.0%
	7 months - 1 year	11	15.7%	5	31.3%	16	18.6%
	Greater than 1 year	18	30.5%	9	64.3%	27	35.5%
	Never tested	8	28.6%	2	66.7%	10	29.4%
Location of Last Test							
	Government facility	73	51.8%	24	75.0%	98	55.7%
	Mobile counseling	22	16.5%	9	40.9%	32	20.1%
	NGO	38	22.5%	15	39.5%	53	25.4%
	Private clinic	12	35.3%	1	100.0%	13	36.1%
	School	0	0.0%	0	0.0%	0	0.0%
	Work	0	0.0%	0	0.0%	0	0.0%
	Other	5	25.0%	1	20.0%	6	24.0%

Appendix C: Select Variables by MSMW and MSM

Table 19c: Various Indicators of MSMW Compared to MSM

		MSM with >=1 Female Sexual Partner (n=205)		MSM with 0 Female Sexual Partner (n=330)	
		n	%	n	%
Sexual Orientation	Bisexual	177	86.3%	100	30.3%
	Gay	23	11.2%	226	68.5%
	Heterosexual/Straight	5	2.4%	3	0.9%
Sexual Position Preference	Receptive	29	14.1%	105	31.8%
	Insertive	74	36.1%	60	18.2%
	Versatile	96	46.8%	151	45.8%
	Oral Only	5	2.4%	3	0.9%
	None	0	0.0%	1	0.3%
Parish of Residence	Missing	1	0.5%	0	0.0%
	Clarendon	15	7.3%	21	6.4%
	Hanover	5	2.4%	8	2.4%
	Manchester	1	0.5%	2	0.6%
	Portland	2	1.0%	3	0.9%
	St Andrew	58	28.3%	141	42.7%
	St Ann	18	8.8%	15	4.5%
	St Catherine	28	13.7%	43	13.0%
	St Elizabeth	1	0.5%	2	0.6%
	St James	47	22.9%	56	17.0%
	St Mary	4	2.0%	5	1.5%
	St Thomas	6	2.9%	6	1.8%
Age Group 1	Trelawney	0	0.0%	4	1.2%
	Westmoreland	20	9.8%	24	7.3%
Age Group 2	16-17	14	6.8%	15	4.5%
	18-24	106	51.7%	158	47.9%
	25+	85	41.5%	157	47.6%
Marital Status	16-17	14	6.8%	15	4.5%
	18-24	106	51.7%	158	47.9%
	25-29	47	22.9%	82	24.8%
	30-39	26	12.7%	49	14.8%
	40-49	9	4.4%	20	6.1%
	50+	3	1.5%	6	1.8%
Condom Use At Last Sex with Female Partner	Married to a man	52	25.4%	145	43.9%
	Married to a woman	37	18.0%	0	0.0%
	Formerly married to a man	39	19.0%	79	23.9%
	Formerly married to a woman	10	4.9%	3	0.9%
	never married	67	32.7%	103	31.2%
Where meets partners (multiple response)	Yes	80	39.0%	N/A	
	No	44	21.5%		
	Declined	1	0.5%		
	Did not list female partner in partner matrix	80	39.0%		
Paid for Sex In Last 12 Months	Hotel	14	6.8%	25	7.6%
	Street	107	52.2%	150	45.5%
	Friends	108	52.7%	194	58.8%
	Intemet	140	68.3%	277	83.9%
	Party	60	29.3%	100	30.3%
	Work	46	22.4%	87	26.4%
	Workshop	21	10.2%	58	17.6%
	Church	19	9.3%	44	13.3%
	Gas Station	13	6.3%	21	6.4%
	Bar	29	14.1%	45	13.6%
	Lime	0	0.0%	69	20.9%
	Gym	6	2.9%	7	2.1%
	School	29	14.1%	67	20.3%
	Motel	6	2.9%	4	1.2%
	Other	11	5.4%	12	3.6%
Reveived Payment for Sex	Yes	41	20.0%	24	7.3%
	No	164	80.0%	306	92.7%
Income	Yes	65	31.7%	69	20.9%
	No	140	68.3%	261	79.1%
Revealed MSM Status at Last Healthcare Visit	40k or less	80	39.0%	129	39.1%
	41-149k	62	30.2%	107	32.4%
	150-299k	20	9.8%	27	8.2%
	300-450k	5	2.4%	11	3.3%
	450+k	4	2.0%	7	2.1%
Revealed MSM Status at Last Healthcare Visit	Missing	34	16.6%	49	14.8%
	Yes	141	68.8%	274	83.0%
	No	52	25.4%	40	12.1%
	Don't know	1	0.5%	0	0.0%
	Did not visit healthcare	11	5.4%	16	4.8%

	MSM with >=1 Female Sexual Partner (n=205)			MSM with 0 Female Sexual Partner (n=330)			
	Mean	Median	IQR	Mean	Median (IQR)	Mean	Median (IQR)
Number of Male Partners	4.95	3	1 to 5	4.58	3.00 (2.00-5.00)	4.38	3.00 (2 to 5)